

Multi-Objective Evolutionary Hyperparameter Optimisation for Recursive Battery Degradation Forecasting

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ABSTRACT

Reliable lithium-ion battery degradation forecasting must balance predictive accuracy with model size and computational demand. These competing requirements motivate multi-objective hyperparameter optimisation, but comparisons between stochastic search methods can be confounded by optimiser variability and neural-training noise. Recursive battery-degradation forecasting was used as a testbed to compare NSGA-II, NSGA-III, and matched-budget random search over validation error, parameter count, and inference latency. Fifteen independent runs per method, with 280 evaluations per run, produced 12,600 complete LSTM trainings on 228 commercial lithium-ion cells. NSGA-II achieved a final mean hypervolume of 1.1972, exceeding NSGA-III (1.1579; Holm-corrected $p = 2.7 \times 10^{-4}$) and random search (1.1410; Holm-corrected $p = 1.2 \times 10^{-5}$). It reached random search's final mean hypervolume after 71.2 evaluations on average in all 15 runs, whereas only 8/15 random-search runs reached the same threshold. Fixed-configuration retraining produced macro-MAPE standard deviations of 0.21–0.52 percentage points, motivating multi-seed re-screening and confirmation. The selected NSGA-II model used 92.5% fewer parameters than the manual reference, while its 1.810% versus 1.856% macro-MAPE difference was not statistically significant. Teacher-forced and recursive ablations produced different feature rankings, while an Arrhenius-derived stress feature worsened prediction on all ten matched seeds, consistent with redundancy and the discontinuity introduced when zero operating-cycle counts are clipped before the logarithm. Under frozen-representation, head-only adaptation, source pretraining produced significant negative transfer on 40 external LG M50T cells. NSGA-II's principal benefit was therefore faster, more repeatable convergence toward useful error-complexity-latency trade-offs rather than a statistically established gain in final predictive accuracy.

1. Introduction

Lithium-ion batteries are central to electric mobility, portable electronics, and electrochemical energy-storage systems because of their high energy density, efficiency, and cycle life. Their performance nevertheless changes progressively during operation. Repeated cycling and calendar ageing induce coupled electrochemical and mechanical processes, including solid-electrolyte-interphase growth, loss of lithium inventory, active-material degradation, and particle cracking, which ultimately appear at the cell level as capacity loss and increasing impedance [Edge et al.(2021)Edge, O'Kane, Prosser, Kirkaldy, Patel, Hales, Ghosh, Ai, Chen, Yang, Li, Pang, Bravo Diaz, Tomaszewska, These changes determine the state of health (SOH) of a cell and directly affect usable energy, power capability, lifetime, and replacement decisions. Reliable estimation and forecasting of battery health are therefore important components of battery management and lifetime planning [Ng et al.(2020)Ng, Zhao, Yan, Conduit, and Seh].

Battery-health modelling has consequently developed along physics-based, equivalent-circuit, statistical, and data-driven directions. Physics-based models can provide mechanistic descriptions of degradation but require parameters that are difficult to identify consistently across cells and operating conditions. Data-driven methods instead learn relationships between measured ageing variables and battery-health indicators and have become increasingly practical as larger experimental ageing datasets have become available [Ng et al.(2020)Ng, Zhao, Yan, Conduit, and Seh, Luh and Blank(2024)]. The Luh and Blank dataset, for example, provides capacity and impedance measurements for

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228 commercial NMC/C-SiO cells aged under different temperatures and calendar, cyclic, and driving-profile conditions [Luh and Blank(2024)]. Such datasets make it possible to examine degradation prediction across heterogeneous operating conditions rather than on a small number of nominally similar cells.

Deep neural networks are particularly useful when degradation measurements form ordered trajectories. Long short-term memory (LSTM) networks were designed to retain information over extended sequences [Hochreiter and Schmidhuber] and have subsequently been applied to battery remaining-useful-life and SOH prediction [Zhang et al.(2018)Zhang, Xiong, He, and]. More recent battery-health studies have combined recurrent networks with convolutional, wavelet, attention, and other learning components to improve feature extraction and temporal modelling [Yao et al.(2024)Yao, Song, and Xie]. These developments have established sequence learning as an effective approach to battery-health prediction, but the performance of a neural sequence model depends not only on its model family. Hyperparameters such as hidden width, network depth, sequence length, dropout, learning rate, and training configuration determine the representation capacity, computational cost, and behaviour of the resulting predictor.

Hyperparameter selection is therefore an optimisation problem in its own right. Manual tuning is difficult to reproduce systematically, while exhaustive grid search becomes expensive as the number of design variables grows. Random search provides a stronger baseline because it allocates trials without requiring uniform exploration of every hyperparameter dimension [Bergstra and Bengio(2012)]. Evolutionary search offers a different advantage: it maintains and modifies populations of candidate solutions and can search simultaneously over discrete and continuous design variables. This has motivated its use for automated neural architecture and hyperparameter selection. Recent studies in *Applied Soft Computing*, for example, have used evolutionary search to construct neural architectures while accounting for predictive performance and model complexity [Garcia-Garcia et al.(2023)Garcia-Garcia, Morales-Reyes, and Escalante], and related work has considered several objectives during neural-network design and pruning [Galván and Stapleton(2023)].

For a deployable forecasting model, however, minimising prediction error alone is not sufficient. A configuration with slightly lower error may require substantially more parameters or slower inference, whereas a compact configuration may provide a more useful compromise at essentially the same predictive accuracy. The model-selection problem is therefore naturally multi-objective. NSGA-II addresses such problems through nondominated sorting, elitist survival, and crowding-distance-based diversity preservation [Deb et al.(2002)Deb, Pratap, Agarwal, and Meyarivan]. NSGA-III retains nondominated sorting but uses reference points to guide population diversity and was developed particularly for problems with several competing objectives [Deb and Jain(2014)]. Both algorithms return approximation sets rather than a single optimum, allowing prediction error, model size, and computational cost to be considered without imposing an arbitrary scalar weighting beforehand. Their performance can consequently be assessed using Pareto-set indicators such as hypervolume and inverted generational distance rather than only the smallest error found during a search [Zitzler et al.(2003)Zitzler, Thiele, Laumanns, Fonseca, and da Fonseca, Ishibuchi et al.(2015)Ishibuchi, Masuda, Tanigaki, and Nojima].

Metaheuristic optimisation has also entered battery-health modelling. Multi-objective NSGA-II has been used to balance SOH-estimation error against the cost of acquiring health features [Cai et al.(2020)Cai, Meng, Stroe, Peng, Luo, and Teo] while multi-objective grey-wolf optimisation has addressed prediction error, feature selection, and symbolic-model complexity [Li et al.(2024)Li, Jin, Li, Yu, Zhang, and Zhang]. Prediction-oriented battery studies have additionally used particle-swarm optimisation for LSTM configuration [Gong et al.(2022)Gong, Zhang, Gao, Li, Yan, Peng, and Huang], Bayesian optimisation with Hyperband for BiLSTM tuning [Yang and Chen(2025)], and swarm-based hyperparameter optimisation of hybrid temporal models [Quan et al.(2025)Quan, Zhou, Li, and Wan]. Related recent work has also coupled population-based optimisers with LSTM models for battery-health estimation [Peng et al.(2025)Peng, Wang, Tang, Jiang, Li et al.(2026)Li, Yan, Zhu, Zhou, and Jia]. These studies demonstrate the usefulness of search-based model configuration. Absolute prediction-error values reported across battery studies are not directly comparable when dataset composition, prediction targets, and forecasting protocols differ. The present study evaluates joint capacity and electrolyte-resistance forecasting across 228 cells under recursive rollout, rather than treating results obtained from smaller or differently formulated prediction tasks as directly comparable benchmarks. These studies do not, however, by themselves establish whether an apparent optimiser advantage remains when the search is repeated under the same computational budget.

This distinction is important because both parts of the optimisation pipeline are stochastic. Evolutionary algorithms vary with population initialization, selection, crossover, and mutation, while neural-network training varies with weight initialization and optimisation trajectory. A result obtained from one optimisation run can therefore reflect a favourable search realization, a favourable neural-training seed, or both. Methodological studies of stochastic optimisation recommend repeated independent runs and statistical comparisons rather

than conclusions drawn from individual best solutions [Derrac et al.(2011)Derrac, García, Molina, and Herrera, LaTorre et al.(2021)LaTorre, Molina, Osaba, Poyatos, Del Ser, and Herrera]. This consideration becomes especially important when candidate neural configurations differ by only small fractions of a percentage point in prediction error.

The evaluation protocol introduces a second source of potential discrepancy for sequential battery models. During teacher-forced inference, measured target history is supplied to the model at successive prediction steps. This setting is useful for controlled one-step evaluation, but it does not represent a long-horizon forecast in which future health measurements are unavailable. Under recursive rollout, the model must instead use its own previous predictions to continue the degradation trajectory. Errors can then propagate across subsequent steps, and hyperparameters or input features that appear unimportant under teacher forcing may become influential once measured target history is removed. Recursive rollout was therefore treated as the principal forecasting condition, with teacher forcing retained only as a controlled comparison.

Against this background, multi-objective evolutionary hyperparameter optimisation was investigated for recursive battery-degradation forecasting. NSGA-II, NSGA-III, and matched-budget random search were evaluated using the same LSTM search space, data partitions, training pipeline, recursive evaluator, and budget of 280 candidate evaluations per run. Three objectives were optimised simultaneously: validation macro mean absolute percentage error (MAPE), trainable parameter count, and inference latency. Each optimisation method was repeated independently 15 times, producing 12,600 complete neural-network training evaluations. Search-stage candidates were subsequently re-screened, and the selected configurations were evaluated over ten training seeds so that optimiser behaviour could be separated from seed-dependent neural-training variation.

Four findings define the contribution of the study. First, under the matched evaluation budget, NSGA-II reached the final mean hypervolume of random search after 71.2 evaluations on average and reached that threshold in all 15 runs, showing that its principal advantage lay in front convergence rather than in the lowest isolated validation error. Second, repeated training of fixed configurations produced macro-MAPE standard deviations of 0.21–0.52 percentage points, demonstrating that single-run candidate rankings can be comparable in magnitude to ordinary training-seed variation. Third, feature rankings changed between teacher-forced and recursive evaluation, and the Arrhenius-derived stress descriptor worsened recursive MAPE on all ten matched seeds. Fourth, reuse of the learned source representation under frozen-representation, head-only adaptation produced negative transfer on an independent LG M50T 21700-cell dataset [Kirkaldy et al.(2024)Kirkaldy, Samieian, Offer, Marinescu, and Patel]. Together, these findings separate search quality, final-model accuracy, deployment-aligned forecasting behaviour, and cross-dataset transportability.

2. Problem Formulation and Data

2.1. Primary Battery Ageing Dataset

The primary data source was the publicly available battery ageing dataset reported by Luh and Blank [Luh and Blank(2024)]. The dataset contains commercial NMC/C–SiO cylindrical cells in the 18650 format and includes capacity and impedance-related measurements collected during repeated health checkups. A total of 228 cells and 3,980 processed checkup records were retained for the present analysis. The cells were aged under a range of thermal and operating conditions, providing heterogeneous degradation trajectories for model development and hyperparameter optimisation.

The nominal ageing temperatures were 0, 10, 25, and 40 °C. Charge rates ranged from 0 to 1.67 C, while discharge rates ranged from 0 to 1 C. The processed capacity values ranged from 1.512790 to 2.970551 Ah, and the electrolyte-resistance measurements ranged from 12.990333 to 28.150000 mΩ. Beginning-of-life capacity values occupied a much narrower interval of 2.926306–2.970551 Ah, while beginning-of-life electrolyte resistance ranged from 12.990333 to 19.766333 mΩ. The resistance target remained well separated from zero throughout the processed dataset, ensuring that percentage-error metrics could be evaluated without near-zero denominators.

Measurements were available at discrete health-check intervals rather than at every ageing step. The observations associated with each cell were therefore treated as an ordered degradation trajectory rather than as independent tabular samples. Cell identity was preserved throughout preprocessing and sequence construction so that records from different cells were never combined within the same temporal sequence.

2.2. External Transfer Dataset

An independent ageing dataset reported by Kirkaldy et al. [Kirkaldy et al.(2024)Kirkaldy, Samieian, Offer, Marinescu, and Patel] was used for the external transfer audit. The dataset contains commercial LG M50T cylindrical cells in the 21700 format and was kept completely separate from the primary Luh–Blank data during model development, hyperparameter

optimisation, and finalist selection. This separation allowed the transfer experiment to test whether a representation learned from the source dataset remained useful when applied to a different cell format and ageing-data distribution.

A subset of 40 cells was retained for the transfer analysis. After the trajectory-wise calibration and evaluation split, the external test portion contained 246 later-half evaluation points. Each cell trajectory was ordered according to the archived ageing index before splitting. The first portion of each trajectory was reserved for target-domain calibration, while the remaining records were used exclusively for evaluation. No external evaluation record was used during the optimisation of the source-domain models.

The external dataset was not used as a direct extension of the primary training corpus. Instead, it supported a controlled transfer experiment in which the source-trained neural representation was frozen and only the prediction head was adapted using the calibration portion of each target trajectory. Its performance was compared with a target-only LSTM trained from scratch under the same target-domain information constraint and with deterministic analytical controls. This protocol isolates the value of representation reuse rather than conflating transfer with full-network retraining.

2.3. Input Features and Prediction Targets

Each source-domain observation was represented by nine exogenous variables describing ageing progression, operating conditions, beginning-of-life cell characteristics, and ageing-protocol information. For cell c at observation index k , the input vector was defined as

$$\mathbf{x}_{c,k} = [k_{\text{exp}}, T, Q_0, R_{e,0}, R_{\text{pulse},0}, C_{\text{chg}}, C_{\text{dis}}, S_{\text{SOC}}, A_{\text{type}}]^T, \quad (1)$$

where k_{exp} denotes the experimental ageing index, T is the ageing temperature, Q_0 is the beginning-of-life capacity, and $R_{e,0}$ is the beginning-of-life electrolyte resistance. The term $R_{\text{pulse},0}$ denotes a beginning-of-life pulse-resistance proxy. In the archived implementation this quantity was stored under the field name `Rct0`; however, it was constructed from the difference between the 1 s and 10 ms pulse-resistance measurements and is therefore not interpreted as an electrochemically fitted charge-transfer resistance. The remaining variables are the charge and discharge C-rates, the calendar-ageing SOC setpoint descriptor, and the ageing-type descriptor. The variable k_{exp} was derived cell-wise from the source field `num_cycles_op` as $k_{\text{exp},i} = \max(\text{num_cycles_op}_i - \text{num_cycles_op}_{\text{first}}, 0)$. The official data structure defines `num_cycles_op` as the number of cycles performed under the applicable operating-ageing conditions. It is not an equivalent-full-cycle count, cumulative-throughput measure, elapsed time, or checkup index. The descriptor S_{SOC} corresponds to the source field `age_soc`: 0 denotes that an SOC setpoint is not applicable for non-calendar cyclic or profile ageing, whereas 10, 50, 90, and 100 denote calendar-ageing SOC setpoints of 10%, 50%, 90%, and 100%, respectively. The descriptor A_{type} corresponds to `age_type`, with 1 denoting calendar ageing, 2 cyclic ageing, and 3 profile ageing.

The source-domain forecasting task was formulated as a two-output regression problem. The target vector for cell c at ageing observation k was

$$\mathbf{y}_{c,k} = [Q_{c,k}, R_{e,c,k}]^T, \quad (2)$$

where $Q_{c,k}$ is the measured discharge capacity and $R_{e,c,k}$ is the electrolyte-resistance measurement associated with the same health-check record. Capacity was retained as the principal health variable because it directly represents the available charge-storage capability of the cell, while electrolyte resistance provided a complementary indicator of internal ageing.

The two targets differ substantially in physical units and numerical scale. Model training was therefore performed on scaled target variables, while all reported prediction errors were calculated only after transforming the outputs back to their original physical units. For multi-output comparison, the target-wise percentage errors were combined through macro MAPE, giving capacity and electrolyte resistance equal weight despite their different units.

The nine variables in Eq. (1) are exogenous inputs and should be distinguished from the target-history channels used by the recurrent models. During sequence-model training, previous capacity and electrolyte-resistance values may additionally enter the recurrent input window. Under recursive deployment, future measured target values are unavailable and are replaced by the model's own preceding predictions. The precise sequence construction and recursive forecasting formulation are defined separately in Section 2.5.

Table 1

Cell-level partition of the sequence-eligible primary dataset.

Subset	Cells	Records	Role
Training	209	3,461	Model fitting
Validation	9	243	Early stopping and model selection
Test	10	276	Final held-out evaluation
Total	228	3,980	—

2.4. Data Preparation, Partitioning, and Leakage Control

The raw ageing records were converted into cell-wise degradation trajectories before sequence construction. Capacity was obtained from the reported aged-capacity estimate, and records with non-positive or physically implausible capacity values below 0.1 Ah were excluded. Within each cell, observations were ordered according to the source ageing record and timestamp, with the latest timestamp retained when duplicate records were available for the same ageing point.

Electrolyte resistance, R_e , was calculated from the available reference-impedance measurements, with the 10 ms pulse-resistance measurement used as a fallback when the impedance-derived value was unavailable. Missing values within an otherwise valid trajectory were interpolated within cell. The beginning-of-life pulse-resistance proxy was derived from the difference between the 1 s and 10 ms pulse-resistance measurements. For cells without a valid pulse-resistance difference, the preserved preprocessing procedure assigned $R_{\text{pulse},0} = 0.35R_{e,0}$. This variable is treated only as a pulse-timescale resistance proxy and not as a fitted electrochemical charge-transfer resistance.

Cells without sufficient valid observations to support the required recurrent input and prediction windows were excluded before model training. Remaining rows containing non-finite or unresolved missing values were also removed. The final sequence-eligible source dataset contained 228 cells and 3,980 observations.

The cells were partitioned before sequence generation so that all observations from a given cell belonged to one and only one subset. The final split contained 209 training cells, 9 validation cells, and 10 held-out test cells, corresponding to 3,461, 243, and 276 processed records, respectively. The validation subset was used for early stopping, hyperparameter evaluation, and model selection, whereas the held-out test subset was reserved until configuration selection had been completed. Table 1 summarises the partition.

Feature and target transformations were fitted exclusively on the training partition and then applied unchanged to validation and test cells. The two prediction targets were scaled in the same training-only manner for neural-network optimisation, while all reported errors were calculated after inverse transformation to the original physical units.

Leakage control was applied at both the data and sequence levels. No recurrent sequence was permitted to cross a cell boundary, no scaler was fitted using validation or test observations, and held-out test trajectories were not used during hyperparameter search or early stopping. Randomness associated with Python, NumPy, PyTorch CPU, PyTorch CUDA, and shuffled data loading was explicitly seeded so that repeated experiments reflected prescribed seed variation rather than uncontrolled data-order variation.

The final optimisation study used the archived processed dataset containing 3,980 observations from 228 cells. Extended preprocessing provenance, source-field mappings, and implementation details are provided in the Supplementary Material.

2.5. Recursive Forecasting and Common Evaluation Window

Battery degradation forecasting was formulated as a sequential multi-step regression problem. For cell c , let $\mathbf{x}_{c,k}$ denote the nine-variable exogenous input vector defined in Eq. (1), and let

$$\mathbf{y}_{c,k} = [\mathcal{Q}_{c,k}, R_{e,c,k}]^T \quad (3)$$

denote the corresponding degradation target. Sequence models operate on an ordered window of L observations rather than on a single checkpoint record.

During supervised sequence training, measured target history is available and can be appended to the exogenous variables. The recurrent input at observation k can therefore be written as

$$\mathbf{u}_{c,k}^{\text{TF}} = [\mathbf{x}_{c,k}, \mathbf{y}_{c,k-1}], \quad (4)$$

where the superscript TF denotes teacher forcing. Under this setting, the measured capacity and electrolyte resistance from the preceding observation are supplied to the predictor.

The deployment-oriented setting differs after the initial observed history has been provided. Future measured targets are assumed to be unavailable, so the previously predicted target is returned to the model as part of the next recurrent input:

$$\mathbf{u}_{c,k}^{\text{RO}} = [\mathbf{x}_{c,k}, \hat{\mathbf{y}}_{c,k-1}], \quad (5)$$

where $\hat{\mathbf{y}}_{c,k-1}$ is the preceding model prediction. The resulting recursive forecast is expressed as

$$\hat{\mathbf{y}}_{c,k} = f_{\theta}(\mathbf{x}_{c,k-L+1:k}, \hat{\mathbf{y}}_{c,k-L:k-1}), \quad (6)$$

with $f_{\theta}(\cdot)$ denoting the trained sequence predictor. Once recursive forecasting begins, measured future values of Q and R_e are not inserted into the prediction history. Any forecast error can therefore propagate into subsequent steps, making this setting more representative of long-horizon deployment than teacher-forced evaluation.

The two evaluation modes are illustrated in Figure 1.

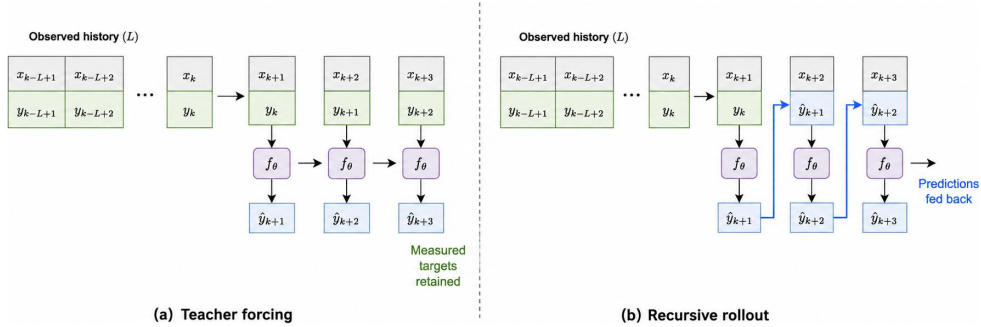


Figure 1: Teacher-forced and recursive rollout evaluation. Under teacher forcing, measured target history is supplied at successive prediction steps. Under recursive rollout, measured future targets are withheld after the initial history window and each predicted capacity–resistance pair is fed back into the subsequent input sequence.

The sequence length L and rollout horizon H were treated as hyperparameters rather than fixed design choices. The search considered

$$L \in \{10, 15, 20\}, \quad (7)$$

and

$$H \in \{3, 5, 8, 10, 15\}. \quad (8)$$

The longest eligible training trajectory contained 29 observations. A candidate was therefore considered feasible only when

$$L + H \leq 29. \quad (9)$$

This coupled constraint is important because a rollout-training segment requires both the initial L -observation history and the subsequent H prediction steps to exist within the same cell trajectory. An earlier manual setting of $L = 20$ and $H = 10$ violated this condition and consequently produced no valid rollout-training segments. The corrected manual reference uses $L = 20$ and $H = 8$.

Direct comparison of different sequence lengths introduces another issue. A model with $L = 10$ can begin producing forecasts earlier in a trajectory than a model with $L = 20$, so evaluating every configuration from its earliest possible prediction would give different models different scoring sets. A common evaluation window was therefore imposed across all candidate sequence lengths. Scoring begins at trajectory row index 20 for every configuration, corresponding to the largest candidate sequence length. Consequently, all optimisation methods and all sequence-length choices are compared on exactly the same observations.

After application of this common-window rule, the validation set contains 63 rollout evaluation points and the held-out test set contains 76 evaluation points. Hyperparameter optimisation and early stopping use only the common validation window. The 76-point held-out window remains untouched until the final configurations have been selected. This common-window design prevents an apparent optimiser advantage from arising simply because one sequence length was evaluated on an easier or shorter portion of the degradation trajectories.

2.6. Evaluation Metrics

Prediction performance was evaluated after transforming the network outputs back to their original physical units. Capacity errors were therefore computed in ampere-hours and electrolyte-resistance errors in m Ω , while percentage-based measures were evaluated from the corresponding unscaled target values.

For target $j \in \{Q, R_e\}$, mean absolute percentage error (MAPE) was defined as

$$\text{MAPE}_j = \frac{100}{N} \sum_{i=1}^N \left| \frac{y_{i,j} - \hat{y}_{i,j}}{y_{i,j}} \right|, \quad (10)$$

where N is the number of evaluated observations, $y_{i,j}$ is the measured value, and $\hat{y}_{i,j}$ is the corresponding prediction. MAPE provides a dimensionless relative-error measure, although its behaviour becomes problematic when target values approach zero [Hyndman and Koehler(2006)]. This limitation does not affect the present source-domain evaluation because both retained targets remain separated from zero.

Since capacity and electrolyte resistance have different physical units and numerical scales, the principal predictive metric was the unweighted mean of their target-specific MAPEs,

$$\text{MAPE}_{\text{macro}} = \frac{1}{2} \left(\text{MAPE}_Q + \text{MAPE}_{R_e} \right). \quad (11)$$

This macro aggregation gives the two degradation targets equal contribution to model selection. RMSE and R^2 were retained as complementary reporting metrics, while the external transfer analysis was evaluated for capacity only because the resistance quantities in the source and external datasets were obtained using different measurement procedures.

Model efficiency was represented by trainable parameter count and measured inference latency. Parameter count was obtained directly from the trainable tensors of each candidate network. Search-stage latency was measured as the median batch-one forward-pass time after 20 untimed warm-up passes followed by five timed blocks. Parameter count and latency were retained as separate objectives because a smaller model does not necessarily execute proportionally faster on a given platform [Phan and Luong(2024), Tan et al.(2019)Tan, Chen, Pang, Vasudevan, Sandler, Howard, and Le].

Search-stage latency should not be confused with the full-trajectory latency reported for the final confirmed models. The former measures one batch-one forward pass and is used as an optimisation objective; the latter includes every recursive forward call required to forecast a complete held-out trajectory.

3. Proposed Multi-Objective Optimisation Framework

The optimisation framework was used to search for recurrent forecasting configurations that balanced predictive accuracy, model complexity, and inference cost under the recursive evaluation protocol defined in Section 2.5. All

Table 2

Structure of the candidate LSTM forecasting model.

Component	Setting	Status
Input channels	9 exogenous + 2 target-history channels	Fixed
Sequence length L	10, 15, or 20	Optimised
LSTM hidden width	64, 96, 128, 160, or 192	Optimised
Number of LSTM layers	1, 2, or 3	Optimised
LSTM dropout	0.2 for depth > 1; otherwise 0	Fixed rule
Dense layer 1	Hidden width $\rightarrow$ 128, ReLU	Fixed
Dropout	0.1	Fixed
Dense layer 2	128 $\rightarrow$ 64, ReLU	Fixed
Output layer	64 $\rightarrow$ 2	Fixed
Rollout horizon H	3, 5, 8, 10, or 15	Optimised
Learning rate η	10^{-4} to 2×10^{-3}	Optimised

candidate configurations used the same forecasting backbone and training procedure. The optimisation algorithms therefore differed only in how candidate hyperparameter configurations were generated and retained, allowing their search behaviour to be compared under matched computational conditions.

3.1. Candidate LSTM Forecasting Model

The forecasting backbone was a stacked long short-term memory (LSTM) network [Hochreiter and Schmidhuber(1997)]. LSTM was selected because the degradation records form ordered cell-specific trajectories and the recurrent state provides a direct mechanism for retaining information from preceding health observations. The architecture itself was kept deliberately compact so that the optimisation study focuses on sequence length, recurrent capacity, depth, rollout horizon, and learning rate rather than on a large number of architecture-specific choices.

At each observation step, the recurrent input contained the nine exogenous variables defined in Eq. (1) together with capacity and electrolyte-resistance history. The resulting per-step input therefore contained 11 variables,

$$\mathbf{u}_{c,k} = \left[\mathbf{x}_{c,k}, Q_{c,k-1}^*, R_{e,c,k-1}^* \right]^T, \quad (12)$$

where the superscript $*$ indicates that the history value depends on the evaluation mode. During teacher-forced training, Q^* and R_e^* were measured values; during recursive rollout, they were replaced by the model's preceding predictions after the observed history window.

For a sequence of length L , the recurrent encoder produced hidden states $\mathbf{h}_{1:L} = \text{LSTM}_\theta(\mathbf{u}_{1:L})$. Only the final recurrent state was passed to the regression head. The head mapped the candidate-dependent hidden dimension through fixed 128- and 64-unit ReLU layers to the two scaled outputs $\hat{Q}$ and $\hat{R}_e$, with dropout 0.1 between the dense layers. A recurrent dropout rate of 0.2 was used when the LSTM contained more than one layer; no recurrent dropout was applied to a single-layer candidate.

Table 2 separates the hyperparameters exposed to optimisation from the architectural components held fixed across all candidates.

Training was performed in two stages. The first stage provided a stable supervised initialization using measured target history. AdamW [Loshchilov and Hutter(2019)] was used with candidate learning rate η , weight decay 10^{-5} , batch size 512, Smooth-L1 prediction loss in scaled target space, and gradient-norm clipping at 1.0. Validation macro MAPE governed early stopping, and the checkpoint with the lowest validation error was restored before rollout fine-tuning.

The second stage adapted the network specifically to recursive prediction. Each rollout-training sample contained an initial window of L observations followed by H future exogenous observations and corresponding targets. Starting from the measured initial history, the first future target pair was predicted, concatenated with the known exogenous variables at the next step, and returned to the recurrent input. This process was repeated for all H steps, with the rollout loss averaged across the prediction horizon.

Rollout fine-tuning used AdamW with learning rate 0.30η , weight decay 10^{-5} , batch size 128, and gradient-norm clipping at 1.0. Early stopping was based on common-window validation rollout macro MAPE rather than teacher-forced error, and the best rollout checkpoint was restored. The two-stage procedure was applied identically to every candidate generated by NSGA-II, NSGA-III, or random search.

3.2. Multi-Objective Search Problem and Search Space

Hyperparameter selection was formulated as a constrained three-objective optimisation problem. Each candidate configuration specified the temporal context available to the recurrent model, its representational capacity, the depth of the recurrent stack, the rollout horizon used during recursive fine-tuning, and the learning rate. The objective was not to identify a single configuration with minimum prediction error, but to obtain a set of nondominated configurations representing different compromises between predictive accuracy, model size, and inference cost.

A candidate configuration was represented as

$$\omega = [L, d_h, n_{\text{LSTM}}, H, \eta], \quad (13)$$

where L is the input sequence length, d_h is the LSTM hidden width, n_{LSTM} is the number of recurrent layers, H is the recursive rollout horizon, and η is the initial learning rate. The corresponding search domains were

$$L \in \{10, 15, 20\}, \quad (14)$$

$$d_h \in \{64, 96, 128, 160, 192\}, \quad (15)$$

$$n_{\text{LSTM}} \in \{1, 2, 3\}, \quad (16)$$

$$H \in \{3, 5, 8, 10, 15\}, \quad (17)$$

$$\eta \in [10^{-4}, 2 \times 10^{-3}], \quad (18)$$

with η sampled and mutated on a logarithmic scale. The resulting search space is mixed discrete–continuous and includes the coupled feasibility constraint

$$L + H \leq 29, \quad (19)$$

which follows from the length of the longest sequence available for rollout training. Candidate configurations violating Eq. (19) are repaired during evolutionary search rather than discarded.

The three optimisation objectives were all minimised. For candidate ω , the objective vector was defined as

$$\mathbf{F}(\omega) = [f_1(\omega), f_2(\omega), f_3(\omega)]^T, \quad (20)$$

where

$$f_1(\omega) = \text{MAPE}_{\text{macro, val}}^{\text{rollout}}, \quad (21)$$

$$f_2(\omega) = P(\omega), \quad (22)$$

$$f_3(\omega) = T_{\text{search}}(\omega). \quad (23)$$

The first objective was the common-window validation macro MAPE obtained under recursive rollout. It therefore evaluates a candidate using the same deployment-oriented prediction setting used for final model assessment. The second objective was trainable parameter count, and the third objective was the median batch-one forward-pass latency measured using the common timing protocol. Model complexity and latency were retained as separate objectives because parameter count is hardware-independent, whereas observed execution time also depends on the computational structure of the network and the execution platform [Tan et al.(2019)Tan, Chen, Pang, Vasudevan, Sandler, Howard, and Le, Phan and Luong(2024)].

Table 3

Hyperparameter search space used by NSGA-II, NSGA-III, and matched-budget random search.

Variable	Symbol	Domain	Type
Sequence length	L	{10, 15, 20}	Discrete
Hidden width	d_h	{64, 96, 128, 160, 192}	Discrete
LSTM layers	n_{LSTM}	{1, 2, 3}	Discrete
Rollout horizon	H	{3, 5, 8, 10, 15}	Discrete
Learning rate	η	10^{-4} – 2×10^{-3}	Log-continuous

Constraint: $L + H \leq 29$.

A feasible candidate was considered to dominate another when it was no worse in all three objectives and strictly better in at least one. The nondominated candidates formed an approximation to the Pareto front. This formulation avoided assigning fixed scalar weights to prediction error, parameter count, and latency before optimisation. NSGA-II and NSGA-III are specifically designed to retain such nondominated trade-offs during population-based search [Deb et al.(2002)Deb, Pratap, Agarwal, and Meyarivan, Deb and Jain(2014)].

Random search was retained as a matched-budget non-evolutionary reference. It samples the same discrete domains and the same log-uniform learning-rate interval, applies the same feasibility condition, and evaluates candidates through the identical training and validation pipeline. Random search is a relevant baseline for hyperparameter optimisation because it does not impose a population-selection mechanism and can remain competitive when only a subset of the available hyperparameters strongly influences model performance [Bergstra and Bengio(2012)]. Consequently, differences between the three search methods can be attributed to candidate-generation and selection behaviour rather than to differences in search space, model training, or objective computation.

The complete optimisation process is summarised in Figure 2.

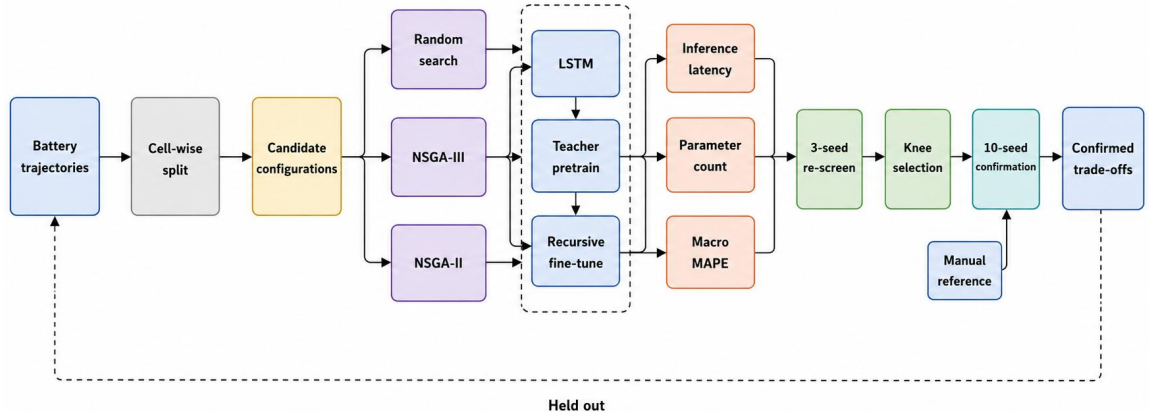


Figure 2: Multi-objective hyperparameter optimisation framework. Candidate LSTM configurations are generated within the constrained mixed search space, trained using the common two-stage forecasting procedure, and evaluated under recursive rollout using validation macro MAPE, trainable parameter count, and batch-one inference latency. NSGA-II, NSGA-III, and matched-budget random search operate on the same candidate representation and evaluation pipeline. Selected nondominated candidates subsequently undergo multi-seed re-screening and final confirmation.

Table 3 summarises the decision variables and their domains.

3.3. Evolutionary Search Procedures

NSGA-II and NSGA-III were implemented over the same candidate representation, feasibility rule, variation operators, objective functions, population size, and evaluation pipeline. Their comparison therefore isolates the effect of the environmental-selection mechanism: crowding-distance preservation in NSGA-II and reference-direction niching in NSGA-III [Deb et al.(2002)Deb, Pratap, Agarwal, and Meyarivan, Deb and Jain(2014)].

Parent configurations were recombined using uniform per-variable crossover. Mutation was applied independently to L , d_h , n_{LSTM} , and H with probability 0.25 per variable. The learning rate was mutated separately with probability 0.25 by Gaussian perturbation in base-10 logarithmic space with standard deviation 0.28 and was clipped to 10^{-4} – 2×10^{-3} . When none of the variables changed, one decision variable was forced to mutate so that the offspring differed from its inherited parent.

The coupled constraint $L + H \leq 29$ was handled through repair rather than candidate rejection. An infeasible offspring retained its generated sequence length while the rollout horizon was replaced by the nearest admissible value; when two feasible horizons were equally distant, the larger value was retained. Duplicate configurations were regenerated and did not consume the unique-evaluation budget.

NSGA-II used binary-tournament parent selection based on Pareto rank and crowding distance. Parent and offspring populations were combined, nondominated sorting was applied, and complete fronts were retained until the population limit was reached. When the final admissible front could only be included partially, candidates with larger crowding distance were preferred [Deb et al.(2002)Deb, Pratap, Agarwal, and Meyarivan].

NSGA-III retained the same nondominated sorting but replaced crowding-distance truncation with reference-direction niching [Deb and Jain(2014)]. For the three-objective problem, a Das–Dennis simplex construction [Das and Dennis(1998)] with division parameter $p = 6$ produced 28 reference directions, matching the population size. Objectives were min–max normalised within the combined parent–offspring population before reference-direction association. Split-front candidates were then selected preferentially from the least-occupied reference niches.

No algorithm-specific change was made to the candidate representation, neural-network training schedule, objective computation, feasibility repair, or unique-evaluation rule. The resulting comparison therefore examines whether the two diversity-preservation mechanisms differ in their ability to discover useful error–complexity–latency trade-offs.

3.4. Matched-Budget Random Search

Random search was included as a non-evolutionary reference for determining whether the population-based mechanisms of NSGA-II and NSGA-III provide an advantage beyond direct sampling of the same hyperparameter space. It is a relevant baseline because it can remain competitive when performance depends strongly on only a subset of the available hyperparameters [Bergstra and Bengio(2012)].

Each candidate was drawn from the same mixed discrete–continuous domain defined in Eq. (18). The four discrete variables were sampled uniformly from their admissible sets, while learning rate was sampled uniformly in logarithmic space over 10^{-4} – 2×10^{-3} . Only configurations satisfying $L + H \leq 29$ were accepted, and duplicates within a run were regenerated without consuming the evaluation budget.

Every retained candidate was trained through the same two-stage LSTM procedure and evaluated using the same validation rollout macro MAPE, trainable parameter count, and measured latency objectives. Random search did not use previous objective values to determine which configuration should be generated next. The comparison was otherwise fully matched in search space, feasibility rules, candidate-training schedule, run-wise seeds, timing procedure, and number of unique model evaluations.

3.5. Pareto-Front Quality and Convergence Indicators

The outcome of a multi-objective search is an approximation set rather than a single optimum. The three methods were therefore compared using hypervolume (HV), inverted generational distance (IGD), IGD^+ , leave-one-method-out IGD (LOO-IGD), front cardinality, and Schott spacing. These indicators assess complementary properties of convergence, coverage, and local distribution of nondominated solutions [Zitzler et al.(2003)Zitzler, Thiele, Laumanns, Fonseca, and Dorigo, Ishibuchi et al.(2019)Ishibuchi, Imada, Masuyama, and Nojima, Schott(1995)].

Because macro MAPE, parameter count, and latency occupy different numerical scales, all 12,600 candidate evaluations were first placed in a common transformed objective space,

$$\mathbf{g}(\omega) = [f_1(\omega), \log_{10} f_2(\omega), \log_{10} f_3(\omega)]^T, \quad (24)$$

and each transformed objective was globally min–max normalised using bounds obtained from the complete evaluation pool. The same fixed bounds were used for every optimiser, independent run, and convergence checkpoint so that the objective-space geometry did not change between comparisons.

Hypervolume was calculated in this common normalised minimisation space using the fixed reference point (1.1, 1.1, 1.1). Larger HV therefore indicates a more favourable combination of convergence and objective-space coverage [Zitzler et al.(2003)Zitzler, Thiele, Laumanns, Fonseca, and da Fonseca].

Because the true Pareto front is unknown, the IGD reference set was constructed as the nondominated union of all 12,600 transformed and normalised candidate evaluations; this pooled set contained 28 points. Smaller IGD denotes closer average proximity to that best-known trade-off surface. IGD⁺ was retained as a dominance-aware robustness check because conventional IGD is not Pareto compliant under all geometries [Ishibuchi et al.(2019)Ishibuchi, Imada, Masuyama].

The pooled-reference construction was additionally checked through LOO-IGD. For each method, the alternative reference set was rebuilt using only evaluations produced by the other two search methods, preventing the scored method from contributing to its own reference set. Each leave-one-method-out reference contained 46 points. Front cardinality and Schott spacing were analysed separately to determine whether indicator differences were associated with the number or local distribution of nondominated solutions.

Indicator values were recomputed after every 28 unique candidate evaluations, giving ten common checkpoints from 28 to 280 evaluations. Evaluation count rather than generation number was used as the common computational axis because random search has no generation-based population update.

3.6. Candidate Re-screening and Final Configuration Selection

The configuration returned by a single search evaluation was not treated as final evidence because neural-network training introduces seed-dependent variation in prediction error and latency. A three-stage procedure was therefore used: multi-objective search, three-seed re-screening, and ten-seed final confirmation. The same procedure was applied to NSGA-II, NSGA-III, and random search.

For each search method, all nondominated configurations obtained across the 15 independent runs were first pooled. Candidate solutions were ordered by their normalised distance from the ideal objective vector, and the five highest-ranked unique configurations were retained. This produced a common re-screening set of 15 candidate configurations across the three search methods.

Each retained configuration was then retrained independently using seeds

$$S_{\text{re}} = \{42, 52, 62\}. \quad (25)$$

For candidate i , the re-screened predictive score was the mean recursive validation macro MAPE across the three seeds,

$$\bar{E}_i = \frac{1}{3} \sum_{s \in S_{\text{re}}} E_{i,s}, \quad (26)$$

where $E_{i,s}$ denotes the common-window rollout macro MAPE obtained for seed s . Parameter count remained deterministic for a given architecture, while latency was remeasured during the re-screening stage.

The three decision quantities used for finalist selection were

$$\mathbf{v}_i = [\bar{E}_i, \log_{10} P_i, \log_{10} \bar{T}_i]^T, \quad (27)$$

where P_i is trainable parameter count and $\bar{T}_i$ is the mean measured forward latency across the re-screening runs. Prediction error, log-parameter count, and log-latency were min-max normalised across all 15 re-screened candidates,

$$\tilde{v}_{i,j} = \frac{v_{i,j} - v_{j,\min}}{v_{j,\max} - v_{j,\min}}, \quad j \in \{1, 2, 3\}. \quad (28)$$

The normalised ideal point is therefore

$$\mathbf{v}^* = (0, 0, 0). \quad (29)$$

A resource-aware knee score was calculated as the Euclidean distance from this ideal point,

$$D_i = \|\tilde{\mathbf{v}}_i - \mathbf{v}^*\|_2 = \sqrt{\tilde{v}_{i,1}^2 + \tilde{v}_{i,2}^2 + \tilde{v}_{i,3}^2}. \quad (30)$$

Within each search method, the candidate with the smallest D_i was selected as the finalist. The rule gives equal geometric weight to re-screened prediction error, logarithmic parameter count, and logarithmic latency after normalisation. It is therefore not an accuracy-priority rule. A configuration with slightly larger MAPE can be preferred when the reduction in model size or latency gives a better overall three-objective compromise. This choice was fixed before final ten-seed evaluation so that the final configuration was not selected retrospectively from held-out test performance.

The selected NSGA-II, NSGA-III, and random-search configurations were then compared with the corrected manually tuned reference configuration. Each of the four models was retrained using ten matched seeds,

$$S_{\text{final}} = \{42, 52, 62, 72, 82, 92, 102, 112, 122, 132\}. \quad (31)$$

All four configurations were evaluated on the same 76-point held-out panel comprising ten cells. The final comparison therefore contained 40 independent model trainings. The ten-seed stage was used only for confirmation of configurations selected before test evaluation; it was not used to modify hyperparameters or to return candidates to the search process.

4. Experimental Design

4.1. Search Budget, Independent Runs, and Seed Pairing

Stochastic optimisation algorithms cannot be compared reliably from a single execution because the observed outcome depends on the particular random realisation of the search. Independent repeated runs and statistical comparison of the resulting performance distributions were therefore used throughout the optimiser study [Derrac et al.(2011)Derrac, García, Molina, and Herrera, LaTorre et al.(2021)LaTorre, Molina, Osaba, Poyatos, Del Ser, and H

NSGA-II, NSGA-III, and random search were each executed independently 15 times. Every run was allocated exactly 280 unique candidate evaluations. The resulting search programme contained 4,200 complete candidate evaluations per search method and 12,600 evaluations across the three methods. Each candidate evaluation comprised a complete neural-network training procedure, including teacher-forced pretraining, recursive rollout fine-tuning, validation rollout evaluation, parameter counting, and latency measurement. The computational budget therefore represents trained forecasting models rather than inexpensive analytical fitness-function evaluations.

For the evolutionary methods, the population size was 28. Ten successive evaluation blocks, including the initial population, were used, giving 280 candidate evaluations per run. The same evaluation counts,

$$e \in \{28, 56, 84, \dots, 280\}, \quad (32)$$

were used as common convergence checkpoints for random search. Comparison against the number of completed candidate evaluations rather than generation number gives all three methods the same computational axis.

Run-level random-number-generator seeds were

$$20270001, 20270002, \dots, 20270015. \quad (33)$$

The same run seed was assigned to the corresponding NSGA-II, NSGA-III, and random-search executions. Neural-network training seeds were independently defined as

$$41001, 41002, \dots, 41015, \quad (34)$$

Table 4

Matched search budget and repeated-run design used for optimiser comparison.

Setting	Value
Search methods	NSGA-II, NSGA-III, random search
Independent runs per method	15
Population/reference size	28
Evaluation blocks per run	10
Unique evaluations per run	280
Evaluations per method	4,200
Total search evaluations	12,600
Run RNG seeds	20270001–20270015
Candidate-training seeds	41001–41015
Seed matching	Paired by run index across methods
Convergence checkpoints	28, 56, . . . , 280 evaluations

and were likewise paired by run index across the three search methods. Thus, run r of each search method used the same prescribed search-level and candidate-training seed pair. This pairing reduces avoidable between-method variation and makes the repeated comparison more sensitive to differences in search behaviour.

The archived integrity checks confirmed that every method–run combination contained exactly 280 unique feasible evaluations, with no missing objective values, duplicate run/evaluation identifiers, or candidates violating the constraint $L + H \leq 29$. The 28-candidate initial populations were also recorded for every run before method-specific search updates were applied.

Fifteen independent runs were used for each search method, providing run-level distributions for statistical comparison while retaining the same 280-evaluation neural-training budget for every run. Across the three methods, this design produced 12,600 complete candidate evaluations. Including the search, noise-floor, re-screening, and final-confirmation stages, the complete optimisation programme accumulated 19.06 h of recorded model-training time. The 15-run design therefore balanced repeated-run statistical inference against the computational cost of complete neural-network training and recursive evaluation.

Table 4 summarises the matched experimental budget and seed design.

4.2. Training Schedules and Computational Environment

The same two-stage training procedure described in Section 3.1 was retained throughout hyperparameter search, candidate re-screening, and final confirmation. The stages differed only in the maximum number of training epochs and early-stopping patience. Shorter schedules were used during the large-scale search to control computational cost, while progressively longer schedules were used once the candidate set had been reduced.

Teacher-forced pretraining used AdamW [Loshchilov and Hutter(2019)] with the candidate-specific learning rate η , weight decay 10^{-5} , batch size 512, and Smooth-L1 loss in scaled target space. Gradient norms were clipped at 1.0. Early stopping was governed by validation macro MAPE, and the checkpoint with the lowest validation error was restored before recursive fine-tuning.

Rollout fine-tuning used AdamW with learning rate

$$\eta_{\text{RO}} = 0.30\eta, \quad (35)$$

weight decay 10^{-5} , batch size 128, and the same multi-step rollout loss used in the candidate-training stage. Gradient norms were again clipped at 1.0. Early stopping was based exclusively on common-window validation rollout macro MAPE, and the best rollout checkpoint was restored before objective computation or held-out evaluation.

Table 5 gives the schedules used at the three stages of the optimisation pipeline.

The search schedule was applied to all 12,600 candidate evaluations generated by NSGA-II, NSGA-III, and random search. Candidate re-screening used the longer intermediate schedule for the 15 retained configurations evaluated over three training seeds. The final schedule was applied only to the three optimiser-selected configurations and the manually specified reference model during ten-seed confirmation. Increasing the available training budget in this

Table 5

Training schedules used during search, candidate re-screening, and final confirmation.

Stage	Teacher max.	Teacher patience	Rollout max.	Rollout patience
Search	80	18	15	5
Re-screening	160	30	30	10
Final confirmation	220	40	40	15

manner reduces the computational burden of broad exploration while allowing the final comparisons to use a more permissive convergence schedule.

The primary optimisation experiments were executed on an NVIDIA Tesla T4 GPU using Python 3.12.13 and PyTorch 2.10.0+cu128. Prescribed run seeds controlled Python, NumPy, PyTorch, CUDA, and shuffled data loading.

Search-stage latency was measured on the same Tesla T4 for every candidate. After 20 warm-up passes, five batch-one forward-pass measurements were recorded and their median was used as T_{search} . Thus, all three search methods were evaluated under the same latency-measurement protocol.

4.3. Training-Noise Characterisation

Neural-network training introduces stochastic variation even when architecture, hyperparameters, data partition, and evaluation protocol are held fixed. This source of variation is particularly relevant in hyperparameter optimisation because small differences between candidate objective values may reflect training-seed effects rather than persistent configuration differences. The magnitude of this variation was therefore measured explicitly before interpreting the repeated optimiser comparisons [Derrac et al.(2011)Derrac, García, Molina, and Herrera, LaTorre et al.(2021)LaTorre, Molina, Osaba, Poyatos, Del Ser, and Herrera].

The noise experiment used eight fixed LSTM configurations spanning the search space. Each configuration was trained independently using the ten seeds {42, 52, 62, 72, 82, 92, 102, 112, 122, 132}, producing 80 complete neural-network trainings. No architecture or hyperparameter was changed between seeds for a given configuration. The same search-stage training schedule was used: at most 80 teacher-forced epochs with patience 18 followed by at most 15 rollout epochs with patience 5.

For each configuration, recursive validation macro MAPE and batch-one inference latency were summarised across the ten retrainings using their mean and sample standard deviation. The purpose of this experiment was descriptive rather than model-selective. None of the eight configurations was promoted to the final comparison because of its noise-floor result. Instead, the measured variability provided an empirical scale against which small search-stage accuracy differences could be interpreted and motivated the independent re-screening and ten-seed confirmation stages.

4.4. Statistical Analysis

Statistical analysis was designed around the experimental unit associated with each comparison. Independent optimisation runs were used for comparisons between search methods, matched training seeds were used for comparisons between confirmed forecasting configurations, and paired target cells were used for the external transfer analysis. All hypothesis tests were two-sided, with a significance level of $\alpha = 0.05$. Non-parametric procedures were preferred because the number of repeated runs or seeds was modest and normality of the resulting performance distributions was not assumed [Derrac et al.(2011)Derrac, García, Molina, and Herrera, LaTorre et al.(2021)LaTorre, Molina, Osaba, Poyatos, Del Ser, and Herrera].

For the optimiser comparison, hypervolume, IGD, IGD⁺, leave-one-method-out IGD, front cardinality, and spacing were obtained independently from each of the 15 runs of NSGA-II, NSGA-III, and random search. For each indicator, equality across the three methods was first examined using the Kruskal–Wallis rank test [Kruskal and Wallis(1952)]. When the omnibus comparison was followed by pairwise analysis, two-sided Mann–Whitney U tests were applied [Mann and Whitney(1947)]. The three pairwise p -values within an indicator family were adjusted using Holm’s sequential procedure to control the family-wise error rate [Holm(1979)].

In addition to p -values, pairwise optimiser comparisons were accompanied by the Vargha–Delaney A_{12} measure of stochastic superiority [Vargha and Delaney(2000)]. A value of $A_{12} = 0.5$ indicates no stochastic ordering, whereas values increasingly different from 0.5 indicate stronger separation between the two run distributions.

The direction of interpretation depends on the indicator. For hypervolume, larger values are preferable, so $A_{12} > 0.5$ favours the first method. For IGD, IGD⁺, leave-one-method-out IGD, and spacing, smaller values are preferable;

consequently, $A_{12} < 0.5$ favours the first method. Effect sizes were therefore interpreted together with the optimisation direction rather than by numerical magnitude alone.

The selected NSGA-II, NSGA-III, and random-search configurations were compared with the corrected manual reference using the ten matched training seeds defined in Eq. (31). Because the same seed was used for each configuration within a comparison, seed-wise outcomes formed paired observations. Two-sided Wilcoxon signed-rank tests were therefore used for pairwise comparisons [Wilcoxon(1945)]. Comparisons of the three optimiser-selected configurations against the manual reference were adjusted using Holm correction.

Seed uncertainty was summarised using the sample mean and sample standard deviation across the ten independently trained models. Ninety-five-percent confidence intervals for mean performance and paired mean differences were obtained using 10,000 bootstrap resamples [Efron and Tibshirani(1993)]. Bootstrap resampling was carried out over the seed-level observations rather than over individual prediction rows, preserving the trained model as the unit of replication.

Differences in seed-to-seed dispersion between the confirmed configurations were examined only as a secondary analysis. Levene and Brown–Forsythe tests were used to assess homogeneity of variance. Variance differences were not inferred from standard deviations alone.

For the separate grouped five-fold model comparison, performance ranks were compared across the evaluated model variants using the Friedman test. The five held-out folds formed the repeated blocks. Because only five folds were available, pairwise exact Wilcoxon tests have a minimum attainable non-zero two-sided p -value of 0.0625. Pairwise superiority was therefore not inferred when the omnibus test rejected equality; model ranks and the omnibus result were reported without unsupported pairwise significance claims.

The external transfer experiment produced paired errors for the same 40 target cells under source-pretrained adaptation and the corresponding target-only control. Positive cell-level differences indicate deterioration after source pretraining. The median paired shift was tested using a two-sided Wilcoxon signed-rank test. A 95% confidence interval for the mean cell-level effect was estimated from 10,000 bootstrap resamples of the 40 paired differences. Resampling was performed at the cell level so that repeated observations from a cell were not treated as independent replicates.

Throughout the manuscript, statistical significance was distinguished from practical or engineering relevance. A non-significant difference in predictive error was not interpreted as proof of equivalence, and numerical differences in seed dispersion were not described as established effects unless the corresponding inferential test supported that conclusion.

4.5. Comparative Baselines and Ablation Protocols

Several auxiliary experiments were run after the primary optimisation study to test whether the main conclusions depended on the LSTM architecture, the selected input representation, or the absence of explicit physical constraints. These experiments were not included in the NSGA-II, NSGA-III, or random-search objective evaluations and did not influence finalist selection.

4.5.1. Sequence-Model Baselines

Two additional sequence architectures were evaluated under the same sparse-checkup recursive forecasting setting: a temporal convolutional network (TCN) and a Mamba selective state-space model. The TCN followed the causal residual-convolution formulation commonly used for sequence modelling [Bai et al.(2018)Bai, Kolter, and Koltun], while the Mamba baseline used the selective state-space architecture introduced by Gu and Dao [Gu and Dao(2023)]. Both received the same 11 per-step channels as the recurrent model and were evaluated using seeds 42, 52, 62, 72, and 82 on an NVIDIA Tesla T4.

The TCN used a sequence length of 20, 96 hidden channels, and four causal residual blocks with dilations 1, 2, 4, and 8. It contained 119,234 trainable parameters and was trained with learning rate 7×10^{-4} , weight decay 10^{-4} , batch size 512, a maximum of 180 epochs, and early-stopping patience 35. The Mamba implementation contained 70,722 trainable parameters. Detailed architecture and execution settings for both auxiliary models are provided in the Supplementary Material.

4.5.2. Physics-Informed Baseline

A physics-informed neural-network baseline was included to test whether explicit degradation constraints improved prediction relative to the same neural backbone trained only from data. The general physics-informed learning principle

follows the formulation of Raissi et al. [Raissi et al.(2019)Raissi, Perdikaris, and Karniadakis]. The implemented objective combined the data loss with initial-condition, monotonicity, smoothness, capacity-degradation, and resistance-growth terms. The physics-loss coefficients were adaptively balanced during training rather than held fixed.

The activation-energy parameter was constrained to 40–80 kJ mol⁻¹ and regularised toward a prior centred at 56 kJ mol⁻¹ with scale 7 kJ mol⁻¹. It was therefore treated as a bounded prior-guided model parameter rather than as an independently measured physical quantity. A data-only version of the same predictor was trained without the physics terms, allowing the effect of the added constraints to be examined without changing the underlying prediction backbone.

4.5.3. Teacher-Forced and Recursive Feature Ablation

Feature importance was assessed under both teacher-forced and recursive evaluation because measured target history can compensate for missing exogenous information during teacher forcing. The full nine-variable input representation was used as the reference configuration, after which individual feature groups and compact subsets were evaluated without changing the remaining training procedure. Particular attention was given to $k_{\text{exp}} + T$ and $k_{\text{exp}} + R_{e,0}$ because these compact subsets produced contrasting outcomes under teacher-forced and recursive evaluation.

4.5.4. Arrhenius-Derived Stress Descriptor

A separate matched-seed ablation tested whether an explicitly constructed Arrhenius-derived descriptor added predictive information beyond the original feature set. The descriptor was calculated as

$$S_{\text{Arr}} = \ln \left[\max(k_{\text{exp}}, 10^{-6}) \right] - \frac{E_a}{RT}, \quad (36)$$

where T is absolute temperature, $R = 8.314 \times 10^{-3}$ kJ mol⁻¹ K⁻¹, and E_a is a per-cell activation-energy estimate obtained from the PINN fitting procedure. For non-training or external cells without a cell-specific E_a estimate, the global mean calculated from the 180 phase-one training-cell estimates was used.

The comparison used ten matched seeds. To isolate the effect of adding the stress channel, the larger model containing the descriptor was initialised first. The corresponding model without the stress variable was then formed by deleting that input column and copying all parameters with matching shapes exactly. The paired comparison therefore did not confound the descriptor effect with a different random initialisation.

The activation-energy distribution was estimated from 180 training cells in the separate phase-one analysis; validation and test cells did not contribute to its estimation. The stress-feature experiment was therefore treated as an auxiliary analysis of a model-derived physical descriptor. Its interpretation focused on two properties: k_{exp} and T were already present in the base feature set, and clipping mapped $k_{\text{exp}} = 0$ to $\ln(10^{-6}) = -13.82$, producing a sharp change relative to $k_{\text{exp}} = 1$. Additional provenance and split details are given in the Supplementary Material.

4.6. Grouped Five-Fold Validation Protocol

A grouped five-fold analysis was performed as an additional check on whether the relative behaviour of the forecasting variants depended strongly on the fixed train–validation–test partition used for hyperparameter optimisation. Entire cell trajectories, rather than individual checkup records, formed the grouping unit. This prevented observations from the same cell from appearing in both model-fitting and held-out subsets within a fold.

The outer partition used five-fold grouped cross-validation on the sorted unique cell identifiers. No shuffling, random state, or stratification by temperature, charge rate, discharge rate, or ageing protocol was applied to the outer folds. Within each outer-training partition, 15% of the cells were assigned to an inner validation subset after fold-specific seeded shuffling. Scaling transformations for the base forecasting variables were estimated from the training cells of the corresponding fold and applied without refitting to validation and held-out cells.

Seven predefined forecasting variants were evaluated in each fold: the teacher-forced LSTM, recursive-rollout LSTM, a sparse-input LSTM, two PINN-derived feature models, the data-only PINN predictor, and the physics-constrained PINN. The PINN-derived stress representations used the phase-one training-derived E_a values described in Section 4.5.4; E_a was not refitted independently within each grouped outer fold. These feature variants are therefore interpreted as robustness checks of the archived representation rather than as fold-specific refits of the physical parameter.

For each variant, held-out macro MAPE was calculated separately in all five folds and converted to within-fold ranks. Differences across the seven variants were tested using the Friedman test [Friedman(1937)]. Because only five folds were available, the smallest attainable non-zero two-sided exact Wilcoxon signed-rank p -value is 0.0625. Pairwise superiority was therefore not claimed from these folds. The outer folds were also not stratified by operating condition, and this design choice is retained as a limitation of the grouped analysis.

4.7. External Transfer Audit Protocol

External transfer was examined on the independent LG M50T 21700-cell dataset reported by Kirkaldy et al. [Kirkaldy et al.(2024)Kirkaldy, Samieian, Offer, Marinescu, and Patel]. Forty cells were retained for this audit, producing 246 later-half evaluation points. The external data were not used during source-domain hyperparameter optimisation, finalist selection, or ten-seed confirmation.

Each target trajectory was sorted by the archived ageing index and divided by record position. Approximately the first half of each trajectory formed the calibration segment, while evaluation began strictly after that segment and continued to the final record. This ensured temporal separation between target-domain fitting and evaluation within every retained cell.

The principal transfer condition used a source-trained neural representation whose representation layers were held fixed during target-domain adaptation. Only the prediction head was updated using the calibration portion of the external trajectories. A target-only neural control was trained from scratch using the same target-domain information, and an analytical square-root-fade model was retained as a deterministic reference.

The external neural experiments were repeated over five independent training seeds. Only capacity prediction was used for cross-dataset comparison because the resistance target in the primary dataset and the 0.1 s pulse-resistance quantity in the Kirkaldy dataset are not physically interchangeable. Capacity MAPE was calculated separately for each target cell over its later-half evaluation records.

Statistical inference was performed at the cell level rather than the individual-record level. Paired transfer differences across the 40 cells were tested using the two-sided Wilcoxon signed-rank procedure described in Section 4.4. Ninety-five-percent confidence intervals for the mean paired effect were obtained from 10,000 bootstrap resamples of the cell-level differences.

The transfer audit was deliberately restricted to this adaptation setting. Full-network fine-tuning, partial layer unfreezing, explicit domain adaptation, and other transfer-learning strategies were not evaluated. Any conclusion drawn from this experiment therefore applies only to reuse of the source representation under frozen-representation, head-only adaptation.

5. Results and Discussion

5.1. Training-Noise Floor

Retraining the same LSTM configuration with different random seeds produced measurable variation in recursive validation error. Across the eight fixed configurations, the standard deviation of macro MAPE ranged from 0.21 to 0.52 percentage points, even though the architecture, hyperparameters, data split, and training schedule were unchanged. The corresponding mean macro MAPE values ranged from 1.55% to 3.26%. The variation was not confined to the less accurate configurations. For example, C2 achieved the lowest mean error of 1.55% with an SD of 0.26 pp, whereas C7 obtained a mean of 1.83% but an SD of 0.52 pp. Thus, two configurations with similar mean performance could produce noticeably different single-run outcomes depending on the training seed.

Latency behaved differently. Across the same 80 trainings, the latency SD remained below 0.03 ms for every configuration, with most values below 0.015 ms. The search-stage latency objective was therefore much less sensitive to training seed than predictive error. This difference matters for the three-objective optimisation problem: parameter count is deterministic, latency was nearly stable under repeated training, whereas validation error contained appreciable stochastic variation.

The observed MAPE variation is large relative to many of the differences produced during hyperparameter search. A candidate that appears better by only a few tenths of a percentage point in one training run can therefore change position after retraining. The lowest value encountered during a large search is particularly susceptible to this effect because it is selected from many noisy evaluations.

For this reason, search-stage objective values were treated as evidence for locating promising regions of the error–complexity–latency space rather than as final estimates of model performance. The three-seed re-screening and ten-seed

Table 6

Pareto-front quality after 280 candidate evaluations. Values are mean $\pm$ standard deviation across 15 independent runs. Larger HV and smaller IGD, IGD⁺, and LOO-IGD are preferred.

Method	HV	IGD	IGD ⁺	LOO-IGD
NSGA-II	1.1972 $\pm$ 0.0185	0.0701 $\pm$ 0.0114	0.0405 $\pm$ 0.0105	0.0581 $\pm$ 0.0177
NSGA-III	1.1579 $\pm$ 0.0244	0.0802 $\pm$ 0.0144	0.0595 $\pm$ 0.0126	0.0812 $\pm$ 0.0148
Random search	1.1410 $\pm$ 0.0164	0.0922 $\pm$ 0.0132	0.0715 $\pm$ 0.0091	0.0939 $\pm$ 0.0191

confirmation stages described in Section 3.6 were used to separate persistent configuration differences from favourable single-seed outcomes. The optimiser comparisons that follow are therefore interpreted primarily through repeated-run Pareto-front indicators, while final predictive claims are based on the independent multi-seed confirmation stage.

5.2. Multi-Objective Convergence and Pareto-Front Quality

The three search methods produced different approximation-front quality under the same budget of 280 candidate evaluations per run. At the final checkpoint, NSGA-II attained a mean hypervolume of 1.1972, compared with 1.1579 for NSGA-III and 1.1410 for random search. The corresponding mean IGD values were 0.0701, 0.0802, and 0.0922, respectively. Thus, the ordering of the methods was consistent across both indicators: NSGA-II produced the largest dominated volume and the smallest average distance to the pooled best-known reference front.

Table 6 gives the final indicator distributions across the 15 independent runs.

The overall hypervolume distributions differed across the three methods (Kruskal–Wallis $p = 2.11 \times 10^{-6}$). Pairwise analysis showed that NSGA-II attained significantly higher hypervolume than NSGA-III (Holm-corrected $p = 2.71 \times 10^{-4}$, $A_{12} = 0.911$) and random search ($p = 1.24 \times 10^{-5}$, $A_{12} = 0.996$). The difference between NSGA-III and random search was not statistically significant after correction ($p = 0.089$). The magnitude of A_{12} indicates that the NSGA-II hypervolume exceeded the corresponding value from a randomly selected NSGA-III or random-search run with high probability.

IGD gave a similar, although less complete, separation. The omnibus comparison was significant ($p = 3.66 \times 10^{-4}$). NSGA-II attained significantly lower IGD than random search (Holm-corrected $p = 4.07 \times 10^{-4}$, $A_{12} = 0.089$). The NSGA-II–NSGA-III and NSGA-III–random comparisons both yielded corrected $p = 0.0559$ and therefore did not cross the 0.05 significance threshold. The corresponding raw p -values were 0.0279 and 0.0512, respectively; their identical adjusted values result from Holm’s monotonic step-down correction rather than from duplicated pairwise tests. For these minimisation indicators, $A_{12} < 0.5$ favours the first-listed method.

The pooled IGD reference set requires an additional check because it was formed from nondominated solutions contributed by all three search methods. NSGA-II supplied a large share of that pooled front, so conventional IGD alone could favour a method that also contributed heavily to the reference set. Two alternative calculations were therefore examined.

First, IGD⁺ retained the same ordering and produced stronger separation. The overall difference was significant ($p = 1.63 \times 10^{-6}$). NSGA-II achieved lower IGD⁺ than NSGA-III (Holm-corrected $p = 5.24 \times 10^{-4}$) and random search ($p = 1.24 \times 10^{-5}$). NSGA-III also achieved lower IGD⁺ than random search ($p = 0.0225$).

Second, leave-one-method-out IGD reconstructed the reference set without any solutions contributed by the method being scored. The overall comparison again differed significantly ($p = 1.92 \times 10^{-5}$). NSGA-II retained lower LOO-IGD than NSGA-III ($p = 9.87 \times 10^{-4}$) and random search ($p = 1.21 \times 10^{-4}$). The NSGA-III–random-search difference remained non-significant ($p = 0.0680$).

These two checks show that the observed NSGA-II advantage was not created solely by its contribution to the pooled reference set. The same ordering persisted when a dominance-aware distance measure was used and when each evaluated method was excluded from construction of its own reference front.

The convergence trajectories provide additional context. Mean hypervolume increased from 1.0878 to 1.1972 for NSGA-II, from 1.0623 to 1.1579 for NSGA-III, and from 1.0455 to 1.1410 for random search between the first and final 28-evaluation checkpoints. Separation between the methods was already visible early in the search and persisted as the evaluation budget increased.

The convergence trajectories are shown in Figure 3.

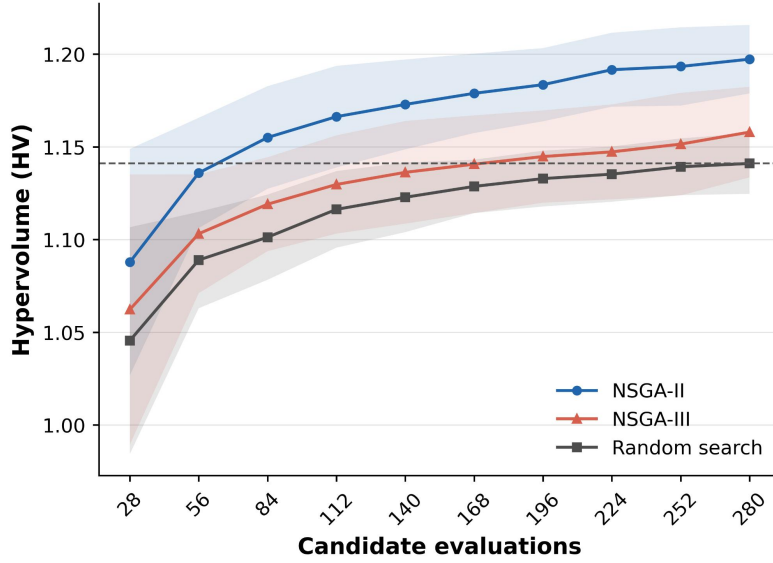


Figure 3: Convergence of NSGA-II, NSGA-III, and matched-budget random search over 280 unique candidate evaluations. Curves show the mean indicator value across 15 independent runs and shaded regions show ± 1 standard deviation. All methods were evaluated at the same 28-evaluation checkpoints using fixed global objective normalisation.

Taken together, the results identify the main search-level difference as proximity to a stronger error–parameter–latency trade-off surface. NSGA-II consistently achieved higher hypervolume than both alternatives and lower distance-based indicators than random search. The comparison with NSGA-III was clearest for hypervolume, IGD⁺, and LOO-IGD, whereas conventional IGD alone did not establish a significant pairwise difference between the two evolutionary methods. This distinction is retained rather than reducing the comparison to a single indicator.

5.3. Evaluation Budget Required to Reach Equivalent Front Quality

Final hypervolume describes approximation-front quality after the complete search budget, but it does not indicate how quickly that level of quality was reached. A complementary analysis therefore measured the number of candidate evaluations required for each run to attain the final mean hypervolume achieved by random search after 280 evaluations. The target hypervolume was fixed at

$$HV_{\text{target}} = 1.141019. \quad (37)$$

For search method m and run r , the first-passage evaluation count was defined as

$$e_{m,r}^* = \min \{ e : HV_{m,r}(e) \geq HV_{\text{target}} \}. \quad (38)$$

Runs that did not attain the threshold within the 280-evaluation budget were treated as non-reachers and were not assigned an artificial first-passage value.

NSGA-II reached the target hypervolume in all 15 independent runs. Among these runs, the mean first-passage budget was 71.2 evaluations, with an SD of 49.8 evaluations. NSGA-III reached the same threshold in 10 of 15 runs and required 125.1 evaluations on average among the successful runs. Random search reached its own final mean hypervolume in only 8 of 15 runs within the allotted budget and required 166.6 evaluations on average among those runs.

Table 7 summarises the first-passage results.

Table 7

Evaluation budget required to reach the final mean hypervolume of random search. Mean and SD are calculated only across runs that reached the threshold; non-reaching runs were retained as censored observations.

Method	Runs reaching target	Mean evaluations	SD	Relative evaluation cost
NSGA-II	15/15	71.2	49.8	0.427
NSGA-III	10/15	125.1	67.0	0.751
Random search	8/15	166.6	68.0	1.000

Among the runs that reached the threshold, NSGA-II required approximately 2.34 times fewer candidate evaluations than random search to attain the same front-quality level. NSGA-III required approximately 1.33 times fewer evaluations than random search among the corresponding successful runs.

The reach rates provide an additional distinction that is not captured by the mean first-passage budget alone. Every NSGA-II run crossed the threshold within the available budget, whereas five NSGA-III runs and seven random-search runs did not. The 2.34-fold comparison should therefore be interpreted only among successful runs rather than as a complete uncensored speed ratio between the methods.

The first-passage budgets and censored runs are visualised in Figure 4.

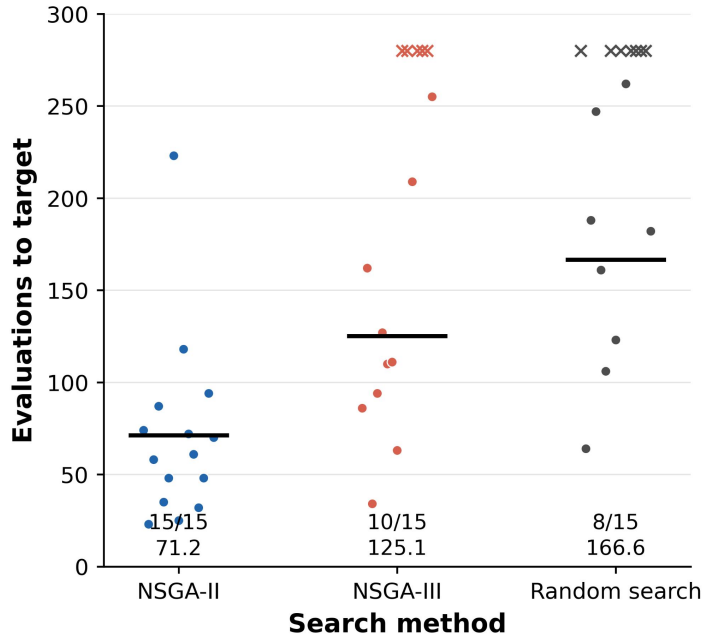


Figure 4: Candidate evaluations required to reach the final mean hypervolume of random search ($HV = 1.141019$). Each point represents one independent search run. Runs that did not reach the threshold within 280 evaluations are shown as censored at the evaluation limit rather than assigned an imputed first-passage value.

The result indicates that the principal benefit of NSGA-II appeared in search efficiency rather than in a large reduction in the best prediction error. It reached a comparable error–parameter–latency trade-off surface using substantially fewer trained candidate models and reached that level in every independent run. This distinction is important for neural hyperparameter optimisation because each candidate evaluation required complete model training and recursive validation.

5.4. Front Cardinality, Spacing, and Pooled Pareto-Front Composition

The improvement in front-quality indicators reported in Section 5.2 could arise for several reasons. An optimiser may return more nondominated solutions, distribute them more evenly across objective space, place them closer to the best-known trade-off surface, or combine these effects. Front cardinality and Schott spacing were therefore examined separately from hypervolume and IGD.

The mean number of nondominated solutions at the final 280-evaluation checkpoint was similar for all three search methods. NSGA-II produced 20.73 ± 5.39 nondominated solutions per run, NSGA-III produced 21.13 ± 4.57 , and random search produced 21.47 ± 4.34 . The Kruskal–Wallis comparison did not indicate a difference in front cardinality ($H = 0.137$, $p = 0.934$).

Schott spacing showed the same pattern. Mean spacing was 0.0840 ± 0.0618 for NSGA-II, 0.0824 ± 0.0499 for NSGA-III, and 0.0703 ± 0.0437 for random search. The overall comparison was not statistically significant ($H = 0.040$, $p = 0.980$). All Holm-adjusted pairwise comparisons for both front cardinality and spacing yielded $p = 1.0$.

The absence of a detectable difference in either quantity narrows the interpretation of the earlier hypervolume result. NSGA-II did not attain higher hypervolume by returning a larger final approximation set, nor was its advantage associated with more uniform local spacing. Instead, its nondominated solutions occupied more favourable regions of the error–parameter–latency objective space and lay closer to the best-known trade-off surface.

A second view was obtained by pooling all 12,600 candidate evaluations and extracting the global nondominated set. This procedure produced 28 solutions: 21 from NSGA-II, five from random search, and two from NSGA-III.

The pooled-front composition does not conflict with the similar per-run front sizes reported above. Each method typically returned approximately 21 nondominated solutions within an individual run, but many of those solutions became dominated when candidates from all methods and runs were compared together. Per-run front indicators and pooled global nondominance quantify different properties: a candidate may belong to a competitive run-level approximation set yet become dominated once the full 12,600-evaluation union is considered. Thus, the smaller pooled contribution of NSGA-III (2/28) than random search (5/28) does not reverse the run-level indicator results; it indicates that fewer NSGA-III candidates survived the stricter global nondominance comparison. NSGA-II accounted for three quarters of the solutions that remained nondominated after this global comparison.

The pooled trade-off structure is shown in pairwise objective projections in Figure 5.

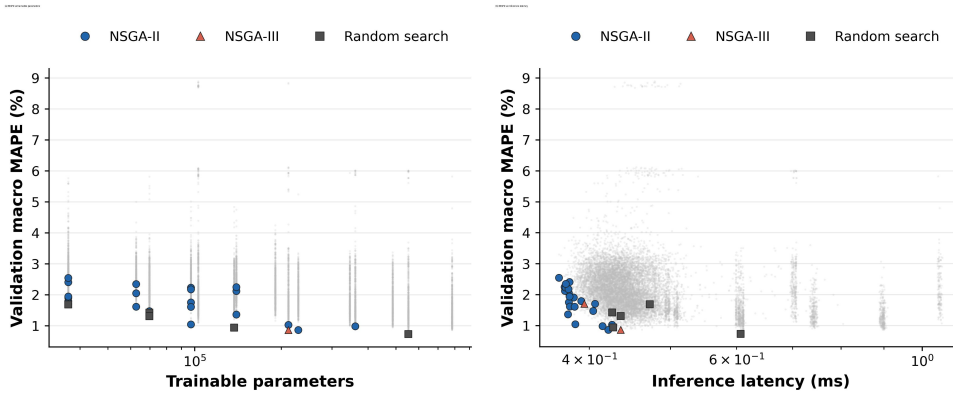


Figure 5: Pooled multi-objective search results across all 12,600 candidate evaluations. Pairwise projections show validation rollout macro MAPE, trainable parameter count, and inference latency in the transformed objective space. The globally nondominated solutions are distinguished by the search method that produced them.

Taken together, the cardinality, spacing, and pooled-front analyses indicate that the NSGA-II advantage was principally one of convergence toward higher-quality trade-offs rather than greater front size or more even within-front diversity. This interpretation is consistent with the higher hypervolume, lower IGD⁺ and leave-one-method-out IGD, and lower evaluation budget required to reach the random-search reference level.

Table 8

Configurations retained for ten-seed final confirmation.

Configuration	L	d_h	Layers	H	Learning rate	Parameters
Manual	20	192	2	8	0.001000000	486,978
NSGA-II	15	64	1	8	0.001215422	36,418
NSGA-III	20	64	1	8	0.000868730	36,418
Random search	15	64	1	5	0.000739324	36,418

5.5. Best-Found Prediction Error and the Limits of Single-Run Winners

The ranking obtained from Pareto-front indicators was not reproduced when each search run was reduced to its single lowest validation macro MAPE. The mean per-run minimum was 0.9518% for NSGA-II, 0.9486% for NSGA-III, and 0.9637% for random search. Their distributions overlapped substantially.

The single lowest search-stage value, 0.7309%, was obtained by random search rather than by either evolutionary method. NSGA-II and NSGA-III both reached minima of 0.8398%. Consequently, a comparison based only on the smallest prediction error observed during optimisation would not support the front-quality ranking reported in Sections 5.2 and 5.3.

This difference arises because the search objectives were explicitly multi-objective. A candidate with the smallest validation error can still be dominated or weakly positioned when parameter count and inference latency are considered jointly. Conversely, a method can produce a stronger approximation front without producing the lowest isolated MAPE value encountered anywhere in the search. The higher hypervolume and lower distance-based indicators obtained by NSGA-II therefore reflect the quality of the complete error–complexity–latency trade-off set rather than superiority of one extreme accuracy-oriented candidate.

The run-wise minima also require caution because each is selected from 280 stochastic neural-network evaluations. Section 5.1 showed that repeated training of a fixed configuration produced macro-MAPE standard deviations of 0.21–0.52 pp. Selecting the minimum from many such noisy evaluations can therefore yield an optimistic estimate of the expected performance of the corresponding configuration.

A direct paired estimate of this selection effect was not available. None of the 45 exact run-winning candidate identifiers appeared in the independently retrained noise-floor or re-screening panels. An exact candidate-matched comparison between search-stage minima and multi-seed retraining was therefore not possible, and no approximate candidate matching was substituted.

The available evidence is instead descriptive. Mean per-run search minima were approximately 0.95%, whereas the ten-seed means of the fixed noise-floor configurations ranged from 1.55% to 3.26%, and the independently re-screened candidate means ranged from approximately 1.95% to 3.51%. These groups contain different configurations and should not be interpreted as a measured “optimism gap”. They nevertheless show why the minimum value observed during a large stochastic search was not treated as the expected performance of the final model.

For this reason, best-found validation MAPE was used only as a descriptive search statistic. Claims about optimiser performance were based on repeated-run Pareto-front indicators, while claims about selected forecasting configurations were deferred to the independent multi-seed confirmation stage.

5.6. Multi-Seed Confirmation of the Selected Configurations

The final comparison did not reproduce the search-level separation between the optimisation methods. After application of the predefined resource-aware knee-selection rule, all three search methods retained the same compact architecture class: a single-layer LSTM with 64 hidden units and 36,418 trainable parameters. The selected configurations differed in sequence length, rollout horizon, and learning rate. As examined further in Section 5.7, this common architecture class was partly a consequence of the selection rule for NSGA-III, whose accuracy-priority re-screened candidate was larger. Consequently, the final-model comparison cannot by itself distinguish the optimisation algorithms; that distinction is supported by the repeated-run Pareto-front results in Sections 5.2–5.4.

The configurations subjected to ten-seed confirmation are listed in Table 8. The manually specified reference contained 486,978 parameters, whereas each optimiser-selected model contained 36,418 parameters.

Across the ten confirmation seeds, the NSGA-II finalist attained the lowest mean macro MAPE at 1.8097%, followed by the manual configuration at 1.8559%, random search at 1.8622%, and NSGA-III at 1.8716%. The

Table 9

Ten-seed held-out performance of the manually specified and optimiser-selected configurations. Values are calculated on the common 76-point recursive test window.

Configuration	Macro MAPE (%)	SD (pp)	95% CI	Q / R_e MAPE (%)
Manual	1.8559	0.2270	[1.7229, 1.9921]	2.8466 / 0.8651
NSGA-II	1.8097	0.3716	[1.5764, 2.0107]	2.8147 / 0.8048
NSGA-III	1.8716	0.3494	[1.6387, 2.0533]	2.6918 / 1.0514
Random search	1.8622	0.1554	[1.7685, 1.9485]	2.8970 / 0.8274

Table 10

Computational characteristics of the confirmed configurations. Values are mean $\pm$ standard deviation across ten training seeds. Forward latency denotes batch-one single-window inference, whereas trajectory latency includes all recursive calls required for one held-out trajectory.

Configuration	Parameters	Forward latency (ms)	Trajectory latency (ms)	Training time (s)
Manual	486,978	0.6956 $\pm$ 0.0014	7.9883 $\pm$ 0.1205	16.482 $\pm$ 2.869
NSGA-II	36,418	0.4248 $\pm$ 0.0202	5.1678 $\pm$ 0.3730	6.227 $\pm$ 0.452
NSGA-III	36,418	0.4219 $\pm$ 0.0158	4.9494 $\pm$ 0.1082	4.690 $\pm$ 0.336
Random search	36,418	0.4130 $\pm$ 0.0175	5.0972 $\pm$ 0.1605	6.461 $\pm$ 0.291

differences were small relative to the seed-to-seed variation. The 95% confidence intervals overlapped substantially for all four configurations.

The mean difference between the NSGA-II finalist and the manual reference was -0.0461 percentage points, with a 95% bootstrap confidence interval of $[-0.3836, 0.2293]$ percentage points. The paired Wilcoxon test gave a raw p -value of 0.7695, and the Holm-adjusted value was 1.0. The corresponding comparisons of the NSGA-III and random-search finalists with the manual configuration were also non-significant. After correction, all three optimiser-selected configurations had $p = 1.0$ for the macro-MAPE comparison.

The numerically lower mean obtained by NSGA-II should therefore not be interpreted as an improvement in predictive accuracy. The ten-seed experiment detected no statistically significant accuracy difference between the NSGA-II finalist and the manually specified reference. The confidence interval was also too wide to establish practical equivalence within a narrow error margin.

The resource differences were considerably larger than the accuracy differences. Each optimiser-selected configuration contained 92.5% fewer trainable parameters than the manual reference. Mean batch-one forward latency decreased from 0.696 ms for the manual model to approximately 0.413–0.425 ms for the compact finalists. Recorded final-training time was also lower for all three selected models.

Table 10 shows the corresponding computational characteristics. Relative to the manual configuration, the NSGA-II finalist reduced batch-one forward-pass latency by 38.9% and recorded training time by 62.2%. Mean recursive-trajectory latency decreased from 7.988 ms for the manual reference to 5.168 ms for NSGA-II, a reduction of 35.3%. The corresponding trajectory latencies were 4.949 ms for NSGA-III and 5.097 ms for random search, representing reductions of 38.0% and 36.2%, respectively. These runtime reductions accompanied a 92.5% decrease in parameter count for all three optimiser-selected models.

Seed-to-seed error dispersion varied numerically across the configurations. Macro-MAPE SD was 0.2270 pp for the manual model, 0.3716 pp for NSGA-II, 0.3494 pp for NSGA-III, and 0.1554 pp for random search. However, neither Levene's test ($p = 0.1316$) nor the Brown–Forsythe test ($p = 0.3134$) rejected equality of variance. The larger numerical dispersion of the NSGA-II and NSGA-III finalists was therefore treated as descriptive rather than as evidence that compact models were less stable.

These results separate two conclusions that would otherwise be easy to conflate. At the search level, NSGA-II converged more consistently toward stronger multi-objective fronts under the matched evaluation budget. At the final-model level, however, the three search methods selected similarly compact LSTM architectures, and no statistically significant improvement in held-out prediction error over the manual reference was detected. The practical gain of the

selected NSGA-II model was therefore primarily a reduction in model size and computational demand rather than a demonstrated increase in predictive accuracy.

5.7. Sensitivity to the Final Selection Rule

The final configuration associated with each search method depended on the decision rule applied after three-seed re-screening. The implemented rule balanced re-screened macro MAPE, parameter count, and latency through the normalised Euclidean distance to the ideal point described in Section 3.6. To examine whether this choice materially affected the reported finalists, the knee-selected configuration was compared with a purely accuracy-prioritised alternative.

For NSGA-II and random search, the two rules selected the same configuration. The lowest re-screened MAPE for NSGA-II was obtained by the $L = 15$, $d_h = 64$, one-layer, $H = 8$ candidate with 36,418 parameters, which was also selected by the implemented knee rule. Random search showed the same agreement for its $L = 15$, $d_h = 64$, one-layer, $H = 5$ candidate.

NSGA-III was the only method for which the selection rule changed the retained configuration. Its lowest re-screened MAPE was obtained by a larger model with $L = 15$, $d_h = 96$, one LSTM layer, and $H = 3$. This candidate contained 62,658 parameters and achieved a three-seed re-screened macro MAPE of $1.9489 \pm 0.1416\%$. The implemented knee rule instead selected the $L = 20$, $d_h = 64$, one-layer, $H = 8$ configuration with 36,418 parameters and a re-screened macro MAPE of $2.2853 \pm 0.0983\%$.

For NSGA-III, the resource-aware rule therefore accepted a re-screened MAPE increase of approximately 0.34 percentage points in exchange for a reduction from 62,658 to 36,418 parameters, corresponding to approximately 41.9% fewer trainable parameters. This was a consequence of the stated three-objective decision rule rather than a failure to identify the lower-error candidate.

The sensitivity analysis also places the final ten-seed comparison in context. The reported NSGA-III finalist in Section 5.6 is the resource-aware knee configuration, not the lowest-error re-screened NSGA-III candidate. The accuracy-priority candidate was evaluated over only the three re-screening seeds and did not undergo the ten-seed final-confirmation experiment. Its 1.9489% result should therefore not be compared directly with the ten-seed means reported in Table 9 as though both estimates had the same level of replication.

The effect of the selection rule was thus limited to one of the three search methods. For NSGA-II and random search, accuracy-priority and multi-objective knee selection coincided. For NSGA-III, the implemented decision rule deliberately favoured the smaller model. This sensitivity should be considered when interpreting the apparent similarity of the three 36,418-parameter finalists: convergence to that common model size was partly a consequence of the resource-aware post-search decision rule rather than evidence that all three search processes identified exactly the same accuracy-oriented solution.

5.8. Teacher-Forced Versus Recursive Feature Ablation

The feature-ablation results depended strongly on the forecasting protocol. Under teacher-forced evaluation, several compact feature subsets appeared to outperform the full input representation. These conclusions changed once the same variants were evaluated recursively, where predicted target values were fed back into subsequent input windows.

Table 11 reports macro MAPE under both protocols. The full-feature baseline obtained 1.168% under teacher forcing and 1.457% under recursive rollout. Removing temperature increased MAPE to 1.312% under teacher forcing and 2.004% under rollout, while removing the ageing index produced corresponding values of 1.374% and 1.847%.

The strongest contrast occurred for the compact input sets. The combination $k_{\text{exp}} + T$ achieved the lowest teacher-forced MAPE, at 0.613%, but its error increased to 1.938% under recursive rollout, placing it fifth among the seven variants. In contrast, $k_{\text{exp}} + R_{e,0}$ achieved a nearly identical teacher-forced error of 0.616% but retained substantially better recursive performance, with the lowest rollout MAPE of 1.284%.

Adding temperature to this compact representation did not improve recursive forecasting. The $k_{\text{exp}} + T + R_{e,0}$ variant achieved 0.682% under teacher forcing but deteriorated to 2.090% under rollout, the highest recursive error among the tested variants. By comparison, the five-variable subset containing k_{exp} , T , $R_{e,0}$, $R_{\text{pulse},0}$, and Q_0 obtained 1.406% under rollout and ranked second.

The disagreement between the two evaluation protocols can be explained by their different information structures. During teacher forcing, the model repeatedly receives measured target history when forming the next prediction. Errors therefore do not propagate through the subsequent sequence. Under recursive rollout, previous predictions replace the corresponding measured target values, allowing local errors and feature deficiencies to influence later forecast steps.

Table 11

Feature-ablation performance under teacher-forced and recursive-rollout evaluation. Lower macro MAPE is preferred. Rankings were calculated separately within each evaluation protocol.

Feature set	Teacher MAPE (%)	Rank	Rollout MAPE (%)	Rank
Full feature set	1.168	5	1.457	3
No temperature	1.312	6	2.004	6
No ageing index	1.374	7	1.847	4
$k_{\text{exp}} + T$	0.613	1	1.938	5
$k_{\text{exp}} + R_{e,0}$	0.616	2	1.284	1
$k_{\text{exp}} + T + R_{e,0}$	0.682	3	2.090	7
$k_{\text{exp}} + T + R_{e,0} + R_{\text{pulse},0} + Q_0$	0.806	4	1.406	2

A compact feature set that performs well when repeatedly anchored by observations is therefore not necessarily the feature set that best supports autonomous multi-step forecasting.

The $k_{\text{exp}} + R_{e,0}$ result is particularly relevant in this respect. Despite using only two exogenous descriptors in addition to the required target-history channels, it produced the lowest recursive MAPE among the tested ablations. The result indicates that a larger input set was not automatically advantageous under recursive deployment. It does not, however, establish that these two variables constitute a universally optimal battery representation; the finding is specific to the present dataset, prediction targets, and rollout protocol.

The ablation results also clarify why feature selection based exclusively on teacher-forced error can be misleading for recursive degradation forecasting. Under teacher forcing, $k_{\text{exp}} + T$ would have been selected as the best-performing subset. Under the deployment-aligned recursive protocol, the same choice would have produced 1.938% MAPE rather than the 1.284% achieved by $k_{\text{exp}} + R_{e,0}$. Feature-ablation conclusions were therefore evaluated using the same rollout conditions intended for final forecasting rather than inferred from one-step prediction alone.

5.9. Effect of the Arrhenius-Derived Stress Feature

Adding the Arrhenius-derived stress descriptor consistently reduced recursive forecasting accuracy. The four-feature base model obtained a macro MAPE of $2.587 \pm 0.341\%$, whereas the otherwise matched model containing the additional stress channel obtained $3.210 \pm 0.434\%$. The mean deterioration was therefore 0.623 percentage points.

The direction of the effect was consistent across all ten matched training seeds. Macro MAPE increased for every seed after the stress descriptor was added, and the paired Wilcoxon signed-rank test confirmed a statistically significant difference ($p = 0.001953$). Because the two models used matched initialisation, the observed change was not attributable to independently sampled starting weights.

The deterioration was larger for resistance prediction than for capacity prediction. Adding the descriptor increased capacity MAPE by approximately 0.30 percentage points, whereas resistance MAPE increased by approximately 0.95 percentage points. These target-specific differences are descriptive because separate hypothesis tests were not performed for the two target errors.

The negative result can be interpreted from the construction of the descriptor in Eq. (36). Two of its principal quantities, k_{exp} and temperature, were already supplied directly to the forecasting model. Across the 180 phase-one training-cell estimates, fitted activation energy occupied a comparatively narrow range of 51.75 to 61.48 kJ mol⁻¹, with a mean of 55.69 ± 2.86 kJ mol⁻¹. The added channel therefore contained a nonlinear transformation of information that was already largely available to the network rather than an independent measurement.

A second issue arose at zero operating-cycle counts. In the archived 3,980-row table used for this analysis, 1,554 rows had $k_{\text{exp}} = 0$. These consisted primarily of calendar-ageing observations, for which the operating-cycle count remained zero, together with initial records from cyclic and profile-aged cells. For these observations, the logarithmic term was evaluated after clipping the input to 10^{-6} , giving $\ln(10^{-6}) = -13.82$, whereas at $k_{\text{exp}} = 1$ the corresponding logarithmic contribution is zero. The transformation therefore introduced a 13.82-unit discontinuity between zero and one operating cycle. The archived minimum stress value of -38.3374 was reproduced by this clipping operation at 0 °C with the mean fitted activation energy.

Table 12

Recursive forecasting performance of the additional sequence-model baselines. Values are mean $\pm$ standard deviation across five training seeds.

Model	Parameters	Macro MAPE (%)	Q MAPE (%)	R_c MAPE (%)
TCN	119,234	1.9775 ± 0.3574	3.0192 ± 0.4899	0.9357 ± 0.4021
Mamba	70,722	2.9453 ± 0.2082	4.3924 ± 0.4713	1.4981 ± 0.1663

TCN timing protocol: latency was measured on an NVIDIA Tesla T4 after five warm-up recursive rollouts per seed, followed by 100 synchronized end-to-end trajectory measurements for each of the five seeds (500 timed measurements in total). The seed-level trajectory latency was 12.559 ± 0.033 ms, where the standard deviation is calculated across the five seed-level means. Across all 500 timing measurements, latency was 12.559 ± 0.429 ms per trajectory. Timing included input scaling, host-to-device transfer, recursive batch-one forward passes, device-to-host transfer, inverse scaling, output clipping, and recursive feedback; dataset loading, model loading, and metric calculation were excluded.

Table 13

Comparison of the matched data-only and physics-constrained PINN predictors. Values are mean $\pm$ standard deviation across repeated trainings.

Model	Macro MAPE (%)
Data-only PINN backbone	7.560 ± 0.195
Physics-constrained PINN	8.049 ± 0.294

The activation-energy term also differed from an independently measured physical input. It was obtained from a fit to capacity-fade data and was subsequently merged into the sequence representation. The stress channel should therefore be interpreted as a derived, model-based descriptor rather than as an additional measured operating variable.

Taken together, the matched-seed result indicates that physical motivation alone did not make the constructed descriptor useful for recursive prediction. In this formulation, the stress variable added information that was largely redundant with existing inputs while also introducing a sharp boundary transformation. Its inclusion worsened macro MAPE by 0.62 percentage points on every matched seed rather than improving the base representation.

5.10. Sequence-Model and Physics-Informed Baselines

5.10.1. Sequence-Model Baselines

The additional sequence architectures did not yield a lower mean recursive forecasting error than the compact LSTM finalists. Across five training seeds, the TCN obtained a macro MAPE of $1.9775 \pm 0.3574\%$, while the Mamba model obtained $2.9453 \pm 0.2082\%$. Their target-specific errors and parameter counts are reported in Table 12.

For reference, the three optimiser-selected LSTM configurations produced ten-seed macro MAPE values between 1.810% and 1.872% in Section 5.6. The TCN result was therefore closer to the compact LSTM range than the Mamba result, but neither additional architecture produced a lower mean recursive error.

The difference was most pronounced for capacity prediction. The TCN obtained 3.019% capacity MAPE and the Mamba model 4.392%, compared with approximately 2.69–2.90% for the confirmed compact LSTM configurations. Resistance prediction was closer to the compact LSTM range, particularly for the TCN, which obtained 0.936% MAPE.

On the Tesla T4, the TCN required 2.211 ± 0.665 s of recorded training time per seed. Synchronized end-to-end recursive inference required 12.559 ± 0.033 ms per trajectory when variation was calculated across the five seed-level latency means. The complete 500-measurement timing distribution had a standard deviation of 0.429 ms.

The result is specific to the sparse recursive setting evaluated here. The usable trajectories contained at most 29 processed checkup records, and inference required repeated feedback of predicted capacity and resistance. Under these conditions, replacing the LSTM with a causal convolutional model or a selective state-space model did not improve predictive accuracy.

5.10.2. Physics-Informed Baseline

Explicit physics constraints also failed to improve prediction over the corresponding data-only backbone. The data-only PINN predictor obtained a macro MAPE of $7.560 \pm 0.195\%$, whereas adding the monotonicity, smoothness, initial-condition, and degradation-residual terms increased the error to $8.049 \pm 0.294\%$.

Table 13 shows that the addition of the implemented physics terms increased mean macro MAPE by approximately 0.49 percentage points. No statistical test supporting a difference between these two variants was preserved, so the comparison is reported descriptively rather than as a statistically significant deterioration. The result supports the narrower conclusion that the implemented physics constraints did not improve predictive accuracy over the same backbone.

The fitted activation-energy parameter provides additional context. The full physics-constrained model produced a value of approximately 57.5 kJ mol^{-1} . This value lay close to the imposed prior centre of 56 kJ mol^{-1} ; the prior scale was 7 kJ mol^{-1} , and the parameter was constrained to the interval $40\text{--}80 \text{ kJ mol}^{-1}$. Across the 180 phase-one training-cell fits used in subsequent feature construction, activation energy ranged from 51.75 to $61.48 \text{ kJ mol}^{-1}$, with a mean of $55.69 \pm 2.86 \text{ kJ mol}^{-1}$. These values are therefore prior-guided fitted quantities and are not interpreted as independent physical measurements.

The result also agrees with the stress-feature experiment in Section 5.9. In both cases, introducing additional physics-derived structure did not reduce recursive forecasting error. The two experiments test different interventions—one changed the training objective and the other added a derived input—so they should not be treated as equivalent failures. Together, however, they show that physical structure was beneficial only if it added information or constraints that were useful under the actual recursive prediction protocol; physical motivation by itself was insufficient.

The broader comparison therefore favoured the compact recurrent formulation for this dataset. Neither the TCN nor Mamba improved on its recursive prediction error, and the implemented PINN constraints did not improve the matched data-only backbone. These findings do not establish a general ranking of sequence architectures or physics-informed learning methods. They define the outcome under the present sparse-checkup trajectories, feature representation, training procedure, and recursive evaluation protocol.

5.11. Grouped Five-Fold Validation

The grouped five-fold analysis examined whether the relative behaviour of the forecasting variants persisted when entire cells were reassigned across different held-out partitions. The seven variants showed a clear difference in their fold-wise rank distributions. The Friedman test rejected equality of ranks across the models ($\chi^2 = 24.171$, $p = 4.86 \times 10^{-4}$).

The teacher-forced LSTM had the lowest mean rank at 1.40, followed by the recursive-rollout LSTM at 2.60. The two compact PINN-derived feature models occupied intermediate positions, with mean ranks of 3.20 and 4.20, while the sparse LSTM obtained a mean rank of 3.60. The data-only and physics-constrained PINN models ranked sixth and seventh, respectively.

The rank ordering should not be interpreted as evidence of pairwise statistical superiority. With only five paired folds, the smallest attainable non-zero two-sided exact Wilcoxon signed-rank p -value is 0.0625. Consequently, no exact pairwise comparison could reach the conventional 0.05 threshold even before adjustment for multiple comparisons. Holm-corrected pairwise superiority claims are therefore not supported by this experiment.

The comparatively favourable rank of the teacher-forced LSTM should also be interpreted in light of the forecasting protocol. Teacher-forced prediction repeatedly receives measured target history and does not experience recursive accumulation of its own prediction errors. Its first-place mean rank therefore does not establish that teacher forcing is the preferable deployment strategy. For the intended multi-step forecasting setting, the recursive-rollout results remain the deployment-aligned evaluation.

The grouped analysis nevertheless supports two broader observations from the fixed-split experiments. First, recurrent LSTM formulations remained among the highest-ranked approaches when the held-out cell composition changed. Second, adding the implemented physics constraints did not improve the data-only PINN backbone; the physics-constrained PINN obtained the highest, and therefore least favourable, mean rank of 7.00.

These results provide evidence that the principal model-family ordering was not confined to a single fixed cell partition. At the same time, the five-fold design does not provide sufficient inferential resolution for claims about individual model pairs. The grouped experiment is therefore treated as a resampling-based robustness check rather than as a second source of pairwise superiority claims.

5.12. External Transfer Audit

The frozen-representation, head-only adaptation protocol produced negative transfer on the external LG M50T dataset. Across the 40 target cells, source pretraining increased per-cell capacity MAPE by a mean of 6.99 percentage points relative to target-only training from scratch. The median paired increase was 5.03 percentage points, and the

Table 14

Paired external-transfer comparisons on the LG M50T dataset. Positive differences indicate higher capacity MAPE for the first-listed method. Confidence intervals were obtained from 10,000 bootstrap resamples of the paired cell-level effects.

Comparison	n	Mean difference (pp)	95% CI (pp)	p -value
Source transfer – target only	40	+6.988	[5.187, 8.795]	1.6×10^{-9}
Source transfer – square-root fade	40	+6.772	[4.907, 8.808]	$< 1 \times 10^{-9}$

bootstrap 95% confidence interval for the mean paired effect was [5.19, 8.79] percentage points. The paired Wilcoxon signed-rank test gave $W = 25$ and $p = 1.6 \times 10^{-9}$.

The negative-transfer result was consistent against both controls. Relative to the square-root-fade model, source-pretrained head adaptation increased per-cell capacity MAPE by 6.77 percentage points (95% CI: [4.91, 8.81]; $W = 16$, $p < 10^{-9}$).

The cell-level transfer effects are shown in Figure 6.

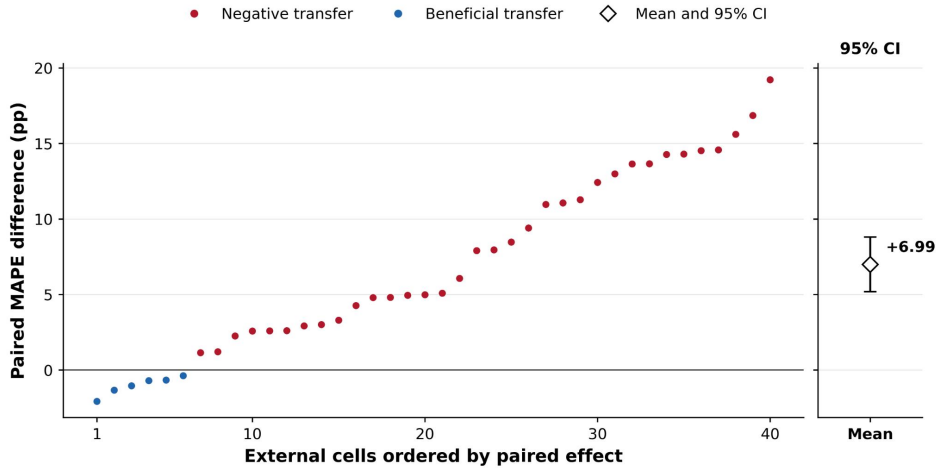


Figure 6: External transfer audit on 40 LG M50T 21700 cells. Cell-level differences compare frozen source-representation, head-only adaptation with target-only training. Positive values denote negative transfer. The summary marker gives the mean paired effect and its 95% bootstrap confidence interval.

The magnitude of the transfer effect was substantially larger than the seed-induced variation observed in the source-domain experiments. The training-noise study produced macro-MAPE standard deviations of 0.21–0.52 percentage points for fixed configurations, whereas the mean external deterioration relative to target-only training was 6.99 percentage points. The external result therefore cannot reasonably be interpreted as a small fluctuation of the same scale as ordinary source-domain retraining variation.

Several differences between the source and target datasets provide plausible context for this result. The optimisation dataset consisted of commercial 18650 cells and contained the source-domain ageing and operating distributions used to learn the frozen representation. The external audit used LG M50T 21700 cells with a different capacity scale and a distinct experimental ageing distribution. In addition, the resistance measurements available in the two datasets were not directly comparable, which is why the external analysis was restricted to capacity prediction. These differences establish a dataset and operating-distribution shift, but they do not by themselves identify the mechanism responsible for the observed negative transfer.

The adaptation strategy also placed a strong restriction on what could be learned from the external calibration data. The source representation was kept fixed, and only the prediction head was updated. A representation whose internal features were specialised to the source trajectories could therefore not reorganise its sequence encoding in response to the target-domain data. By contrast, the target-only model was free to learn its representation directly from the available target calibration records.

The result should consequently be scoped to the protocol that was actually tested. It demonstrates significant negative transfer under *frozen-representation*, *head-only adaptation* from the present source corpus to the evaluated LG M50T trajectories. Full-network fine-tuning, partial unfreezing, representation alignment, adversarial domain adaptation, and other transfer-learning strategies were not evaluated. The experiment therefore does not establish that transfer learning is generally ineffective for battery degradation forecasting.

This external audit also places the source-domain optimisation gains in perspective. Efficient convergence to a strong Pareto front and selection of a compact source-domain model did not guarantee that the learned representation would remain useful after dataset shift. Model efficiency, source-domain predictive accuracy, and cross-domain transportability are therefore distinct properties and require separate evaluation.

5.13. Integrated Interpretation of Optimisation, Forecasting, and Transfer Results

The main effect of multi-objective evolutionary search appeared in the quality and rate of convergence of the approximation front, not in the lowest prediction error encountered during search. Under the same 280-evaluation budget, NSGA-II produced the highest final hypervolume, retained its advantage under IGD⁺ and leave-one-method-out IGD, and reached the final mean hypervolume of random search after 71 evaluations on average in all 15 runs. By contrast, the distributions of run-wise minimum MAPE overlapped, and the lowest single search-stage error was produced by random search. The front-size and spacing results further showed that the advantage did not arise from returning more nondominated solutions or measurably more even local spacing, but from occupying more favourable regions of the three-objective space.

The measured training-noise floor changes how individual search winners should be interpreted. Retraining fixed configurations produced macro-MAPE standard deviations of 0.21–0.52 percentage points, so differences of only a few tenths of a percentage point can alter candidate ordering after retraining. Accordingly, search-stage minima were not treated as final estimates of model quality. The ten-seed confirmation similarly separated search quality from final-model accuracy: the selected NSGA-II model had a numerically lower mean MAPE than the manual reference, but the difference was not statistically significant and the confidence interval did not establish equivalence. Its clearer practical outcome was the 92.5% reduction in parameter count together with lower inference and training cost.

The forecasting experiments also show why evaluation must match intended use. Feature rankings changed substantially between teacher-forced and recursive evaluation, and the feature subset preferred when measured target history was repeatedly supplied was not the subset preferred under autonomous rollout. The Arrhenius-derived stress descriptor increased error on all ten matched seeds, while the implemented physics constraints did not improve the corresponding data-only backbone. These results do not argue against physically informed modelling; they show that physically motivated information or constraints must add useful information under the actual recursive forecasting protocol.

Finally, the external audit places a boundary on the source-domain optimisation results. Under frozen-representation, head-only adaptation, source pretraining increased per-cell capacity MAPE by 6.99 percentage points on the 40 external 21700 cells relative to target-only training. This result does not generalise to full fine-tuning or other domain-adaptation strategies, which were not tested. Taken together, the experiments support a narrower but stronger conclusion than an accuracy-centred optimiser comparison: NSGA-II reached higher-quality error–complexity–latency fronts more quickly and consistently, but that search-level advantage did not translate into a statistically established improvement in final held-out accuracy.

6. Limitations and Future Work

The present findings should be interpreted within the scope of the adopted data, model family, and optimisation setting. The primary search was conducted on 228 commercial lithium-ion cells using a predefined LSTM search space and a fixed budget of 280 candidate evaluations per run. The comparison therefore characterises the relative behaviour of NSGA-II, NSGA-III, and random search under this experimental setting. Broader studies involving additional multi-objective optimisers, larger datasets, and alternative sequence architectures would help determine how consistently the observed convergence advantage persists. Surrogate-based multi-objective Bayesian optimisation provides a complementary strategy for expensive black-box evaluations [Paria et al.(2020)Paria, Kandasamy, and Póczos] and was not examined in the present matched-budget comparison. Its behaviour under noisy recursive neural-network objectives therefore remains a useful direction for further investigation.

A second limitation concerns stochastic candidate evaluation. Each search-stage configuration was trained once within an optimiser run, while the noise-floor experiment showed that repeated training of a fixed configuration can introduce macro-MAPE variation of several tenths of a percentage point. The subsequent re-screening and ten-seed confirmation stages reduced the effect of this uncertainty on final model selection, but the search process itself still operated on noisy objective values. Future work could incorporate adaptive replication or uncertainty-aware objective modelling so that additional training effort is concentrated on candidates close to the evolving Pareto front.

Inference latency was measured under a fixed Tesla-T4 batch-one protocol and is therefore interpreted comparatively within that execution environment. The resource-aware knee rule also affected the NSGA-III finalist, showing that post-search preference articulation can influence which Pareto solution is selected. Alternative decision rules constitute a useful direction for future study.

The implemented physics constraints and Arrhenius-derived descriptor did not improve recursive forecasting under the present formulation, but this outcome does not preclude other forms of physics-guided learning. The activation energy used by the stress descriptor was a fitted, model-derived quantity rather than an independent physical measurement, and it was not re-estimated within each grouped-validation fold. These analyses are therefore interpreted as auxiliary evaluations. The final held-out panel contained 76 rollout points from 10 cells, while the grouped analysis used five folds without operating-condition stratification. Evaluation on larger multi-source cohorts would provide stronger evidence of generalisation across ageing regimes.

Finally, the external audit examined one independent LG M50T 21700 dataset using frozen-representation, head-only adaptation. The observed negative transfer therefore applies to that adaptation strategy rather than to transfer learning in general. Partial unfreezing, full-network fine-tuning, representation alignment, and explicit domain-adaptation methods remain to be tested. Extending the evaluation across additional cell formats and operating conditions would clarify whether the source-domain Pareto-front advantage translates to more transferable forecasting models.

7. Conclusion

A repeated-run multi-objective optimisation study was conducted for recursive battery degradation forecasting, with prediction error, model size, and measured inference latency treated as competing objectives. Under an identical budget of 280 candidate evaluations per run, NSGA-II produced the strongest overall Pareto-front convergence. Its final mean hypervolume was 1.1972, compared with 1.1579 for NSGA-III and 1.1410 for matched-budget random search, and the advantage over both alternatives was statistically significant. NSGA-II also reached the final mean hypervolume of random search after 71.2 evaluations on average and reached this threshold in all 15 independent runs. The corresponding threshold was reached by only 10/15 NSGA-III runs and 8/15 random-search runs.

These gains should not be interpreted as evidence that NSGA-II discovered a more accurate final forecasting model. The lowest individual search-stage MAPE was obtained by random search, while repeated training of fixed configurations showed macro-MAPE standard deviations of 0.21–0.52 percentage points. Independent re-screening and ten-seed confirmation were therefore needed before the selected models were compared. The final NSGA-II model achieved a mean macro MAPE of 1.810%, compared with 1.856% for the manual reference, but the difference was not statistically significant. Its clearer advantage was computational: the selected model contained 92.5% fewer parameters, while the held-out accuracy difference remained statistically non-significant.

The forecasting experiments further showed that conclusions depended on the evaluation protocol. Feature rankings changed between teacher-forced and recursive prediction, and an Arrhenius-derived stress descriptor increased macro MAPE on all ten matched seeds. The implemented physics constraints also did not improve the matched data-only predictor. Finally, frozen- representation, head-only adaptation to 40 external LG M50T 21700 cells produced significant negative transfer, increasing per-cell capacity MAPE by 6.99 percentage points relative to target-only training.

Overall, the principal contribution of multi-objective evolutionary search in this setting was faster and more repeatable convergence toward useful error–complexity–latency trade-offs rather than a statistically established gain in final prediction accuracy. The results also show why stochastic hyperparameter optimisation should be assessed through independent search runs, candidate re-screening, deployment-aligned recursive evaluation, and multi-seed confirmation rather than through a single best training result.

CRediT authorship contribution statement

Simhaa T. T.: Methodology, Software, Formal analysis, Investigation, Data curation, Validation, Visualization, Writing – original draft.

B. Sivakumar: Supervision, Conceptualization, Methodology, Formal analysis, Project administration, Writing – review & editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source-domain Luh–Blank battery ageing dataset is publicly available through RADAR4KIT [Luh and Blank(2024)]. The external LG M50T 21700-cell dataset is publicly available through Zenodo [Kirkaldy et al.(2024)Kirkaldy, Samieian, Offer, and the archived release is available at <https://doi.org/10.5281/zenodo.22057384>. The code associated with the study is available at <https://github.com/Simhaatt/battery-moo-recursive-forecasting>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve language, readability, and organisation. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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